

The observation scheme in the statistics of generalized repunit primes: an exact conditional test of the Lenstra–Pomerance–Wagstaff constant

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Abstract

The Lenstra–Pomerance–Wagstaff (LPW) heuristic predicts $\#\{n \leq x : (b^n - 1)/(b - 1) \text{ prime}\} \sim (e^\gamma / \ln b) \ln x$. Since 1993 it has been tested via a “normalized occurrence frequency” from regressing $\ln n_k$ on k .

First, the suspicion that right-censoring biases this frequency downwards is *false*: both the endpoint form and the OLS slope are exactly unbiased under inverse sampling, and unbiased up to a term exponentially small in μ under a fixed frontier (at most 0.73% for the endpoint form and 2.3% for OLS when $\mu \geq 7$); only the reciprocal is biased—Jensen’s inequality. The genuine defect is that competing forms of the estimator disagree by up to 29.5% on identical data, and the “lucky base” label flips between them.

Second, pooling $B = 15$ bases yields $M = 218$ events against 51 for Mersenne primes alone, but the customary pivot $\kappa S \sim \text{Gamma}(M, 1)$ is exact only under inverse sampling, whereas the data arise from a fixed frontier with a random number of finds, and the frontier itself is not recorded. The unknown frontiers are nuisance parameters, one per base, and the profile estimator M/S is inconsistent in the sense of Neyman and Scott. Our main result is an exact conditional likelihood for the latter scheme (Proposition 3): the position D_b of the last event is sufficient for the frontier, and conditionally on it $N_b - 1$ is *exactly* Poisson($\lambda_b D_b$), a law that does not involve the frontier at all. Hence $M - B \mid D \sim \text{Poisson}(\kappa S)$: the correct pivot under a fixed frontier is Poisson rather than Gamma, the correct pooled estimator is $(M - B)/S$ rather than $(M - 1)/S$ —conditioning costs one event *per base*, as one degree of freedom per group in the Neyman–Scott problem—and confidence intervals are available in closed form without simulation. At the nominal 5% level the old test rejects a true hypothesis in 11–16% of cases depending on the frontier. The revised conclusion is $\hat{\kappa} = 1.838$, exact 95% interval $[1.59, 2.11]$, $p = 0.67$ against $e^\gamma \approx 1.781$; the reported ten-percent excess is an artifact of the scheme.

Third, the inference rests on a single assumption: the frontier was chosen without regard to the finds. If it fails, the estimators differ by 6.9% or more, and the data cannot tell which is right. Without the assumption only an upper bound holds rigorously (Wald’s identity), and a rule such as “stop after a long stretch without finds” biases even $(M - B)/S$ upwards. Four documented frontiers allow the first empirical check of the assumption: they lie closer to the last finds than it predicts, so the dangerous direction is not supported. Hence our call to publish the frontiers L_b . Finally, the exact conditional multinomial homogeneity test ($\Lambda = 7.13$, $p = 0.935$) detects only a spread of κ_b above $\pm 37\%$, the pooled sample only deviations above 21%.

Fourth, the LPW heuristic itself predicts a finite- n intensity $(\kappa / \ln b)(1 + c_b/t)$ with $c_b \in [\ln 2, 1.74]$, because the divisors of $R_b(p)$ are at least $2p + 1$, and the exact conditional inference carries over to this model. The data do not determine c_b , yet it moves the estimate of κ by 8–16%—more than the choice of observation scheme. With it, LPW fits the data ($p = 0.63$ – 0.74 for $c = \ln 2$, $n > 10$), the truncation trend disappears, and the folklore “slow convergence from below” is explained.

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1 Introduction

For an integer base $b \geq 2$ the generalized repunit is

$$R_b(n) = \frac{b^n - 1}{b - 1}, \quad (1)$$

and $R_b(n)$ can be prime only when n is prime. For $b = 2$ these are the Mersenne numbers $M_p = 2^p - 1$. Building on the heuristics of Wagstaff [1] and Pomerance [2], the LPW conjecture asserts that

$$\pi_b(x) = \#\{n \leq x : R_b(n) \text{ prime}\} \sim \frac{e^\gamma}{\ln b} \ln x, \quad (2)$$

where γ is the Euler–Mascheroni constant; a detailed derivation of the heuristic for $b = 2$ is given in the PrimePages reference [3]. The base enters (2) only through the factor $1/\ln b$: the constant $e^\gamma \approx 1.7811$ is declared universal. Dubner [4] proposed testing this universality empirically across bases, and Bourdelais [5] has since maintained running estimates of the normalized frequency, publishing them in the comments to OEIS entries [6]. Related questions for the Mersenne sequence alone were discussed recently in [7]. Close in spirit is also [8], which looks for secondary structure in the Mersenne exponents related to the factorization of $p - 1$; methodologically it is built like our §8—stratified conditional likelihood plus a significance test—but it tests for structure *within* a single base, whereas we test for it *between* bases.

An interpretive folklore has grown up around these estimates. Base 13 is described as “lucky” and base 7 as “the unluckiest”; it is noted that the normalized frequencies lie below the conjectured limit $e^{-\gamma} \approx 0.5615$ and converge to it slowly from below. The present work examines these claims, but arrives at conclusions other than those one would naturally expect.

1.1 Contributions

1. **The frequency estimator in common use is not biased.** The scheme in which the fitting interval ends at the largest known index does indeed condition the estimate on the last observation being an event, and the sufficient statistic is distributed as $\text{Gamma}(N, \lambda_b)$ (Proposition 1). But this does *not* imply that the normalized frequency is biased: both the endpoint form $D/(N \ln b)$ and the OLS slope are exactly unbiased under scheme A, and unbiased up to the factor $1 - (1 + \mu)e^{-\mu}$ under scheme B (Proposition 2, Table 2); at the observed $\mu \approx N_b \geq 7$ the deficit does not exceed 0.73% for the endpoint form and 2.3% for OLS without the origin. Only the reciprocal $\hat{\kappa} = 1/\hat{G}$ is biased, by the factor $N/(N - 1)$, and that is Jensen’s inequality, not censoring. Unbiased estimators of κ and of $1/\kappa$ cannot be reciprocals of one another; any “correction” that repairs one of them damages the other (§3).
2. **The genuine defect is instability with respect to the form of the estimator.** Several numerically distinct variants are in circulation. On identical data they disagree by as much as 29.5% (base 18)—1.7 times the median deviation of the estimates from $e^{-\gamma}$ (17.6%), which is what gets interpreted substantively; for two of the fifteen bases the choice of form weighs more than their entire deviation from the LPW prediction. Under the endpoint form base 18 is the third “unluckiest” of fifteen, and under OLS it is the tenth,

below the median. The label describes the choice of estimator, not a property of the primes (§3.4).

3. **Main result: an exact conditional likelihood for scheme B.** The pivot $\kappa S \sim \text{Gamma}(M, 1)$ is exact under the scheme “observe until the N -th event”, where N is fixed by design. The data, however, arise from a fixed frontier: the search covered $(t_0, \ln L_b]$ and the number of finds is random. The unrecorded window widths W_b are nuisance parameters whose number grows with the number of bases; their joint MLE $\hat{W}_b = D_b$ leads to the inconsistent profile estimator M/S —precisely the Neyman–Scott situation [9]. We show (Proposition 3) that D_b is sufficient for W_b when κ is held fixed and that, conditionally on D_b , the quantity $N_b - 1$ is $\text{Poisson}(\lambda_b D_b)$, with *the window width W_b not entering this distribution*. Hence $M - B \mid D \sim \text{Poisson}(\kappa S)$ is an exact Poisson pivot, and $(M - B)/S$ is the conditional maximum-likelihood estimator in the sense of Andersen [10], exactly conditionally unbiased; it is also obtained by integrating over W_b against the measure dW_b (Remark 6). No moment approximations, no exponentially small remainders and no simulation-based inversion of intervals are required (§4). The distinction between the schemes is itself classical, and the unbiasedness of $(N - 1)/D$ under scheme A is a textbook fact; we lay no claim to either. What is new is that the frontier is *undocumented*, so the exposure must be reconstructed from the finds themselves, and the correction turns out to be proportional to the number of bases (Remark 5).
4. **Numerical consequence.** The exact conditional inference gives $\hat{\kappa} = 1.838$ with a 95% interval $[1.594, 2.109]$ and $p = 0.672$: the excess over LPW disappears. Under scheme B the old Gamma pivot rejects a true hypothesis in 11–16% of cases (§4.4); the spread depends on how far the frontier has advanced, which is precisely why it is reported as a range.
5. **The single assumption and its price.** The whole inference rests on condition (A1): the frontier was chosen without regard to the finds. The estimators $(M - 1)/S$ and $(M - B)/S$ are built on the same statistics (N_b, D_b) and differ by 6.9%; which of them is right is determined not by the data but by how the data were collected. Without (A1) only an *upper* bound holds rigorously: by Wald’s identity $\mathbb{E}[M] \geq \kappa \mathbb{E}[S]$ under any stopping rule. There is no lower bound: the rule “stop the search after a long stretch without finds” biases even $(M - B)/S$ upwards (Remark 9, Table 3). An attempt to sidestep the choice by assembling the estimate from per-base pieces that are unbiased by construction fails (§4.6). Four documented frontiers ($b = 2, 10, 22, 26$) provide the first empirical check of (A1). The frontiers turn out to lie closer to the last finds than (A1) predicts ($p = 0.10$); that is, a deviation, if there is one, points in the safe direction $m < 1$ (§4.5). The only radical remedy is to publish the frontiers L_b .
6. **Lucky bases: what the test cannot do.** Conditioning on $K = M - B$ makes the null distribution of the homogeneity test exact (multinomial), with no appeal to χ^2 asymptotics. We obtain $\Lambda = 7.13$ on 14 degrees of freedom, exact $p = 0.935$. But its power is such that it resolves only a spread of κ_b of order $\pm 37\%$ (§5.3). The correct conclusion is “base-dependent frequencies are not required by the data”, not “they do not exist”.
7. **The second-order term of the heuristic itself.** At finite n the LPW heuristic predicts not a constant intensity but $\lambda_b(t) = (\kappa / \ln b)(1 + c_b/t)$ with $c_b \in [\ln 2, 1.74]$: the divisors of $R_b(p)$ are at least $2p + 1$. The exact conditional inference carries over to this model in closed form (§9). The data do not determine c_b , but the estimate of κ depends on it by 8–16%, that is, more strongly than on the observation scheme. For $c = \ln 2$ and $n > 10$, LPW with its own second-order term is consistent with the data ($p = 0.63$ – 0.74), and the trend of the estimate under truncation disappears. The same term explains the folklore “slow convergence from below”.

8. **An explicit power calculation.** With a two-sided test the pooled sample resolves deviations from e^γ of at least 21.3%; the Mersenne sequence alone, at least 45.5% (§6). The test convention is stated explicitly, since the one-sided variant gives a noticeably different threshold (18.7%).

Section 7 works through a narrower but instructive episode: a naive regression of the Mersenne counting function produces an apparently significant negative intercept, which turns out to be an artifact of conditioning. We include it as a worked example of the same error in a different guise.

2 Model and data

2.1 The Poisson model

Put $t = \ln n$. The heuristic (2) is equivalent to the statement that the indices of generalized repunit primes to base b form a homogeneous Poisson point process on the t -axis with constant intensity

$$\lambda_b = \frac{\kappa}{\ln b}, \quad \text{LPW: } \kappa = e^\gamma = 1.781072\dots \quad (3)$$

Universality is the assertion that a single parameter κ governs all bases. The Poisson idealization here is a working assumption, not a theorem: even for the ordinary primes the Poisson law in short intervals has been established only conditionally, under the Hardy–Littlewood conjectures, and it is known to break down as the scale grows [11]. We adopt (3) as a model and treat κ throughout as a quantity to be estimated; this reduces the question “is LPW true?” to a one-parameter one and makes the bases poolable.

2.2 Admissible bases

Bases that are perfect powers are excluded: for $b = c^m$ with $m > 1$ the number $R_b(n)$ admits an algebraic factorization and the process does not have the assumed form. Bases 4, 8, 9, 16 are therefore omitted. Within the declared range the selection does not depend on the results: we take *all* bases $2 \leq b \leq 20$ that are not perfect powers, which gives $B = 15$; no base was excluded or added on the strength of its estimate. Each of them has at least eight known indices (the minimum is attained at $b = 18$: eight terms of A133857).

We stress, however, that the upper limit $b \leq 20$ is itself a conventional round number and not a consequence of the criterion “at least eight indices”: that criterion continues to hold beyond it (A127996 for $b = 21$ has eight terms, A127997 for $b = 22$ has nine) and first fails at $b = 23$ (A204940, six terms). To make sure the limit does not look chosen after the fact, we tested its influence directly. Extending the set to all bases $b \leq 26$ that are not perfect powers (this adds $b = 21, 22, 23, 24, 26$; $b = 25 = 5^2$ is excluded algebraically) gives $B = 20$, $M = 253$ and

$$\frac{M-1}{S} = 1.9776, \quad \frac{M-B}{S} = 1.8285,$$

that is, it shifts the main estimate (20) by 0.5%—more than an order of magnitude less than the price of the choice of observation scheme (§4). The homogeneity test on the extended set does not reject homogeneity ($\Lambda = 10.24$ on 19 degrees of freedom, exact $p = 0.953$; tag [S17]). No conclusion of this work depends on the limit $b \leq 20$; we retain it as pre-declared.

Remark 1 (Selection on the number of finds). Formally the limit $b \leq 20$ was declared in advance, but it was chosen in a world where the numbers of known terms are already published, and this deserves to be checked not rhetorically but through the sign of the shift. The conditional inference of §4 conditions on $N_b \geq 1$ —which is harmless, since every base has finds; conditioning on $N_b \geq 7$, by contrast, would not be harmless and would bias $\hat{\kappa}$ *upwards*. Lifting the limit to

$b \leq 26$ adds, in particular, the base $b = 23$ with six terms, and shifts the estimate *downwards*, from 1.8381 to 1.8285. The sign of the shift is the one predicted by selection on the number of finds, and its size is an order of magnitude smaller than the price of the observation scheme. We regard this as sufficient evidence that the limit does not drive the conclusion.

2.3 The data and their status

Table 1 lists the sequences used, retrieved on 27 August 2026 (cross-check: Appendix A). Four caveats matter.

Probable primes. For large indices most of the entries are probable primes established by a Fermat or strong Fermat test after trial division. The count is biased upwards only if a Fermat pseudoprime has been mistaken for a prime, which for numbers of this size is practically unheard of. Sensitivity is examined in §5.4.

Verification frontiers. For $b = 2$ the GIMPS project [12] publishes two frontiers: all exponents below $X_{\text{ver}} = 81\,648\,221$ have been tested twice, which gives exactly 50 Mersenne primes, whereas all exponents below $X_{\text{ft}} = 141\,308\,443$ have been tested at least once, which gives 52. The two largest exponents are provisional: unverified candidates remain between them. The analysis of §7, where the frontier is used explicitly, is restricted to the 50 verified exponents. The pooled analysis is insensitive to this choice (§5.4).

Both frontiers are moving quantities, so we fix the moment of reading: the PrimeNet Milestones report, 27 August 2026. No conclusion depends on the exact value of X_{ver} : it suffices that it lies between $M_{50} = 77\,232\,917$ and $M_{51} = 82\,589\,933$, and this holds for every value the report has taken during 2026.

The status of entry A000043. The DATA section of that entry contains 50 terms and stops at 77 232 917: OEIS regards the order of the subsequent exponents as unproven until double-checking is complete. The exponents 82 589 933 and 136 279 841 are known and documented by GIMPS (and are mentioned in the comments to A000043), but formally they are not part of DATA. We include them, because the statistic (13) uses only the position of the last event and not its ordinal number; §5.4 verifies that restricting base 2 to the fifty double-checked exponents does not change the conclusion.

Search bounds and completeness of coverage for $b > 2$. For the other bases a dated and attributed frontier above the last find has been published only for $b = 10$: “Search limit is 10800000, currently” (S. Batalov, comment on A004023, 1 July 2021). In the extended set it is joined by $b = 22$ and $b = 26$, with the bound 10^5 (R. Price, comments on A127997 and A127999, 2012). For the remaining bases the OEIS entries contain only the dates of the finds, and participants in the distributed search state explicitly that the search limits for most bases are unknown. It is precisely this gap that is the subject of §4 and, we believe, the principal obstacle to a quantitative test of LPW; the four documented frontiers are used in §4.5. We note separately that the entire construction assumes that the search covered $(t_0, \ln L_b]$ *contiguously*. If the sieved ranges contain gaps—a common occurrence in distributed searching—then M is understated relative to the exposure and $\hat{\kappa}$ is biased downwards, that is, in the same direction as our main conclusion. In general, completeness of coverage cannot be checked from published data. That such gaps are real is shown by base 5. In 2024 the probable primes $R_5(4\,939\,471)$ and $R_5(5\,154\,509)$ were found (R. Propper); they are not in the OEIS, and summary tables mark the range 3 300 593–4 939 471 as not searched contiguously. For the conditional inference this matters: law (12) requires contiguous coverage of $(t_0, t_0 + D_b)$. We therefore retain $n_{\text{max}} = 3\,300\,593$. Including both numbers, which is admissible only under contiguous coverage, would give $\hat{\kappa} = 1.852$ instead

Table 1. Sequences used. N_b is the number of events strictly to the right of $t_0 = \ln n_{\min}$, that is, one less than the number of known prime indices: the smallest index sets the origin, not an event; $K_b = N_b - 1$ is the conditional event count entering (13). The D_b column is rounded, but all computations use exact $D_b = \ln n_{\max} - \ln n_{\min}$. In the last row $M = \sum_b N_b$, $K = M - B$ and $S = \sum_b D_b / \ln b$; the quantity S is not the sum of the D_b column. Source: OEIS entries, retrieved 27.08.2026.

b	OEIS	primes	N_b	K_b	$n_{\min}$	$n_{\max}$	D_b	$D_b / \ln b$
2	A000043	52	51	50	2	136 279 841	18.0371	26.0220
3	A028491	23	22	21	3	8 530 117	14.8605	13.5266
5	A004061	19	18	17	3	3 300 593	13.9110	8.6434
6	A004062	17	16	15	2	3 360 347	14.3344	8.0002
7	A004063	10	9	8	5	1 264 699	12.4409	6.3934
10	A004023	11	10	9	2	8 177 207	15.2237	6.6116
11	A005808	13	12	11	17	1 868 983	11.6077	4.8408
12	A004064	14	13	12	2	769 543	12.8604	5.1754
13	A016054	13	12	11	5	1 503 503	12.6139	4.9178
14	A006032	11	10	9	3	1 724 417	13.2618	5.0252
15	A006033	10	9	8	3	639 833	12.2704	4.5311
17	A006034	12	11	10	3	1 990 523	13.4053	4.7315
18	A133857	8	7	6	2	1 270 141	13.3615	4.6228
19	A006035	11	10	9	19	209 359	9.3074	3.1610
20	A127995	9	8	7	3	984 349	12.7011	4.2397
total		$M = 218$		$K = 203$		$S = 110.4423$		

of 1.838 (tag [S18]). This is one more argument for publishing not only the frontiers but also the sieving protocols.

3 What is biased and what is not

3.1 The estimators in common use

Dubner [4] and Bourdelais [5] characterize a base b by a normalized occurrence frequency obtained by fitting a straight line to $\ln n_k$ against k and dividing by $\ln b$. By (3) the expected slope is $1/\lambda_b = \ln b/\kappa$, so the normalized frequency estimates

$$G_b = \frac{1}{\kappa}, \quad \text{LPW: } G_b = e^{-\gamma} = 0.5614595 \dots \quad (4)$$

Several variants are in circulation—ordinary OLS with an intercept, regression through the origin, and the endpoint form $\hat{G}_b = D_b/(N_b \ln b)$ with $D_b = t_{N_b} - t_0$. All of them share one structural feature: the fitting interval ends at the largest known index.

3.2 The exact distribution

Proposition 1 (Conditional law of the right endpoint). *Let the events form a homogeneous Poisson process of intensity λ on (t_0, ∞) , and let $t_1 < \dots < t_N$ be its first N events after t_0 . Then $D := t_N - t_0$ has the $\text{Gamma}(N, \lambda)$ distribution with density $\lambda^N d^{N-1} e^{-\lambda d} / (N-1)!$. Conditionally on $D = d$, the preceding $N-1$ events are distributed as the order statistics of $N-1$ independent uniform variables on $(t_0, t_0 + d)$.*

Proof. The waiting times $t_1 - t_0, \dots, t_N - t_{N-1}$ are independent $\text{Exp}(\lambda)$ variables, and a sum of N of them is $\text{Gamma}(N, \lambda)$. Conditional uniformity is the order-statistics property of the Poisson process, applied to $(t_0, t_0 + d)$. $\square$

Corollary 1 (Bias of the reciprocal). *The maximum-likelihood estimator of λ based on D alone is $\hat{\lambda} = N/D$, and*

$$\mathbb{E}[\hat{\lambda}] = N \mathbb{E}[D^{-1}] = \frac{N}{N-1} \lambda, \quad (5)$$

since $\mathbb{E}[D^{-1}] = \lambda/(N-1)$ for $D \sim \text{Gamma}(N, \lambda)$ with $N \geq 2$. The estimator $\tilde{\lambda} = (N-1)/D$ is exactly unbiased.

3.3 What does not follow from Corollary 1

It is natural to conclude that, since $\hat{\lambda}$ is inflated by the factor $N/(N-1)$, the frequency $\hat{G}_b$ must be deflated by the same factor. This is false under either of the two observation schemes analysed below.

Proposition 2 (Unbiasedness of the frequency). (a) Scheme A. *Under the hypotheses of Proposition 1:*

- (i) *the endpoint estimator is exactly unbiased: $\mathbb{E}\left[\frac{D}{N \ln b}\right] = \frac{1}{\lambda \ln b} = G_b$;*
- (ii) *the OLS slope in the regression of t_k on k is exactly unbiased: $\mathbb{E}[\hat{\beta}] = 1/\lambda$;*
- (iii) *consequently the “corrected” quantity $\tilde{G}_b = \frac{N}{N-1} \hat{G}_b$ is biased upwards, by exactly the factor $N/(N-1)$.*

(b) Scheme B. *Let the process be observed on a window of width W , and put $\mu = \lambda W$; the estimators are set to zero where they are undefined. Then*

$$\mathbb{E}\left[\frac{D}{N} \mathbf{1}_{\{N \geq 1\}}\right] = \mathbb{E}[\hat{\beta}_0 \mathbf{1}_{\{N \geq 1\}}] = \frac{1}{\lambda} \left(1 - (1 + \mu)e^{-\mu}\right), \quad (6)$$

where $\hat{\beta}_0$ is the OLS slope through the points (k, t_k) , $k = 0, \dots, N$, including the origin. For the slope $\hat{\beta}_1$ through the points $k = 1, \dots, N$ without the origin, which is defined for $N \geq 2$,

$$\mathbb{E}[\hat{\beta}_1 \mid N \geq 2] = \frac{1}{\lambda} \cdot \frac{1 - (1 + \mu + \mu^2/2)e^{-\mu}}{1 - (1 + \mu)e^{-\mu}}. \quad (7)$$

Both forms are thus unbiased up to a term exponentially small in μ ; at $\mu = 7$ the deficit is 0.73% in (6) and 2.25% in (7), and at $\mu = 10$ it is 0.05% and 0.23%.

Proof. (a)(i) $\mathbb{E}[D] = N/\lambda$ for $D \sim \text{Gamma}(N, \lambda)$, whence the result at once. (ii) The OLS slope is the linear statistic $\hat{\beta} = \sum_k w_k t_k$ with weights $w_k = (k - \bar{k})/\sum_j (j - \bar{k})^2$, for which $\sum_k w_k = 0$ and $\sum_k w_k k = 1$. Since $\mathbb{E}[t_k] = t_0 + k/\lambda$, we get $\mathbb{E}[\hat{\beta}] = \sum_k w_k (t_0 + k/\lambda) = 1/\lambda$. (iii) is immediate from (i).

(b) Under scheme B, conditionally on $N = n \geq 1$, the positions of the events are the order statistics of n independent uniform variables on $(t_0, t_0 + W]$, so $D/W \sim \text{Beta}(n, 1)$ and $\mathbb{E}[D \mid N = n] = Wn/(n+1)$, whence $\mathbb{E}[D/N \mid N = n] = W/(n+1)$. Averaging over $N \sim \text{Pois}(\mu)$ and taking into account that D is defined only for $N \geq 1$,

$$\mathbb{E}\left[\frac{D}{N}\right] = W \sum_{n \geq 1} \frac{\mu^n e^{-\mu}}{n! (n+1)} = \frac{W}{\mu} \sum_{n \geq 1} \frac{\mu^{n+1} e^{-\mu}}{(n+1)!} = \frac{W}{\mu} \left(1 - e^{-\mu} - \mu e^{-\mu}\right),$$

which is (6) for the endpoint form. For the OLS slope, note that $\mathbb{E}[t_k \mid N = n] = t_0 + kW/(n+1)$ for $k = 0, \dots, n$; since the slope is a linear statistic with weights $\sum_k w_k = 0$, $\sum_k w_k k = 1$, in both variants $\mathbb{E}[\hat{\beta} \mid N = n] = W/(n+1)$. For $\hat{\beta}_0$ the summation runs over $n \geq 1$ and gives the same result as above. For $\hat{\beta}_1$ it runs over $n \geq 2$: $\sum_{n \geq 2} \frac{W}{n+1} \frac{\mu^n e^{-\mu}}{n!} = \frac{W}{\mu} (1 - (1 + \mu + \mu^2/2)e^{-\mu})$, and division by $\Pr(N \geq 2) = 1 - (1 + \mu)e^{-\mu}$ gives (7). $\square$

Table 2. Monte Carlo check of Proposition 2, 60 000 replications per row (seed fixed, see `analysis_v2.py`, tag [S1]). The target value of every column except the second is 1.0000. In the right-hand part $\mu = \lambda W$; the column “theory” gives the factor $1 - (1 + \mu)e^{-\mu}$ of (6) for $\hat{G}$. The scheme B column $\mathbb{E}[\hat{\beta}]\lambda$ is for the slope $\hat{\beta}_1$ without the origin, conditional on $N \geq 2$; its theoretical value is given by (7): 0.9775, 0.9977 and 1.0000 at $\mu = 7, 10, 16$.

$N = \mu$	scheme A (N fixed)					scheme B ($\mu = \lambda W$)		
	$\mathbb{E}[\hat{\lambda}]/\lambda$	$N/(N-1)$	$\mathbb{E}[\hat{\lambda}]/\lambda$	$\mathbb{E}[\hat{G}]\lambda$	$\mathbb{E}[\hat{\beta}]\lambda$	$\mathbb{E}[\hat{G}]\lambda$	theory	$\mathbb{E}[\hat{\beta}]\lambda$
7	1.1655	1.1667	0.9990	1.0012	1.0020	0.9908	0.9927	0.9790
10	1.1103	1.1111	0.9992	1.0007	1.0018	0.9997	0.9995	0.9948
16	1.0650	1.0667	0.9985	1.0016	1.0017	0.9992	1.0000	0.9992
22	1.0462	1.0476	0.9987	1.0011	1.0006	0.9991	1.0000	0.9992
51	1.0206	1.0200	1.0006	0.9994	0.9992	0.9997	1.0000	0.9999

The substantive point is simple: unbiased estimators of κ and of $1/\kappa$ cannot be reciprocals of one another, because $x \mapsto 1/x$ is strictly convex. The unbiased pair is $(\hat{G}_b, \tilde{\kappa}_b)$ with $\tilde{\kappa}_b = (N_b - 1) \ln b / D_b$; but $\tilde{\kappa}_b \neq 1/\hat{G}_b$. Any single “censoring correction” that repairs one of these quantities damages the other. Table 2 confirms Proposition 2 by simulation under both schemes.

Remark 2. It is essential that part (b) removes a possible objection to the first conclusion of this work. Below (§4) we adopt scheme B as matching the way the data were obtained; if part (a) were the only one, the whole of §3 would rest on a scheme that the paper itself rejects. Part (b) shows that the conclusion does not depend on the choice of scheme: the deficit $(1 + \mu)e^{-\mu}$ at $\mu \approx N_b$ is 0.73% for $N_b = 7$, 0.05% for $N_b = 10$ and about $2 \cdot 10^{-6}$ for $N_b = 16$ (for OLS without the origin, (7): 2.25%, 0.23% and $1.4 \cdot 10^{-5}$), whereas the deviations discussed in the literature are tens of percent.

Remark 3. The negative result of this subsection deserves emphasis, because it runs against intuition: the systematic downward deviation of the published frequencies from $e^{-\gamma}$ is *not* explained by the fitting scheme. Its source is not the arithmetic of the estimator but the second-order term of the heuristic itself, $c_b \ln \ln x$ in the counting function, which depresses the frequency at finite n and vanishes only as $1/\ln n$ (§9); the observation scheme also contributes to the gap (§4).

3.4 Instability with respect to the form of the estimator

Unbiasedness does not mean agreement. Table 5 gives the endpoint form $\hat{G}_b$ and the OLS estimate $\hat{G}_b^{\text{OLS}}$ on identical data. The discrepancies reach 29.5% ($b = 18$), 15.9% ($b = 13$) and 10.3% ($b = 11$).

Compare these with what the literature interprets substantively—the deviations of the estimates themselves from $e^{-\gamma}$. Across the fifteen bases the latter have median 17.6% (from 5.6% at $b = 20$ to 43.7% at $b = 19$). The largest discrepancy between forms, 29.5%, exceeds that median by a factor of 1.7. On average the choice of form is of course cheaper: the median discrepancy is 4.1%, about a quarter of the median deviation. But for two of the fifteen bases ($b = 18$ and $b = 20$) the choice of form weighs more than their entire deviation from the LPW prediction—and one of them, base 18, is precisely a base the folklore discusses.

The consequence for the folklore is direct. Under the endpoint form base 18 has $\hat{G}_{18} = 0.6604$, the third largest of fifteen (after $b = 7$ with 0.7104 and $b = 10$ with 0.6612), so it belongs firmly among the three “unluckiest”; under OLS, $\hat{G}_{18}^{\text{OLS}} = 0.4656$, tenth of fifteen and below the median 0.4904, so it has become “lucky”. The label flips without a single new prime, purely from a change in the form of the estimator: the base moves from third place to tenth.

To this is added a second, technical difficulty with OLS: its standard errors are wrong in this problem. The residuals are deviations of order statistics and are strongly autocorrelated; for

$b = 2$ we obtain $\rho_1 = 0.825$, and the Newey–West covariance estimator with eight lags raises the standard error of the intercept from 0.447 to 0.732 (§7). Published point values come with no intervals at all, and this—not bias—is the principal methodological objection to current practice.

3.5 The conditional likelihood under scheme A

Proposition 1 yields more than an analysis of bias. Setting $t_0 = \ln n_{\min}$ and taking (N_b, D_b) as the data for base b , we obtain the log-likelihood

$$\ell_A(\kappa) = \sum_b \left[N_b \ln \frac{\kappa}{\ln b} - \frac{\kappa}{\ln b} D_b \right] + \text{const}, \quad (8)$$

whose distribution does not depend on the search bound L_b . This matters in practice: L_b is documented only for $b = 2$ and $b = 10$. Differentiating (8) and putting $S = \sum_b D_b / \ln b$, we get

$$\hat{\kappa} = \frac{M}{S}, \quad \tilde{\kappa} = \frac{M-1}{S}, \quad M = \sum_b N_b. \quad (9)$$

Since $\kappa S = \sum_b \lambda_b D_b$ is a sum of independent $\text{Gamma}(N_b, 1)$ variables,

$$\kappa S \sim \text{Gamma}(M, 1), \quad (10)$$

an exact pivot; confidence intervals and p -values follow in closed form. *It is precisely this statement that requires revision*—see §4.

Remark 4 (Independence across bases). Pooling presupposes that the processes for different bases are independent. Strictly this fails at small indices: seventeen index values occur in more than one base, and $n = 3$ occurs in nine of them, since $R_b(3) = b^2 + b + 1$ is prime for many b . All such coincidences lie at $n \leq 317$. More important than coinciding indices (the numbers $R_b(n)$ for different b are distinct) is the possible sharing of algebraic divisors; both affect only small n , whereas D_b is determined by the upper end of the range. Section 5.4 confirms that discarding all indices below 10^3 leaves the conclusion unchanged. We note that this remark bears more strongly on the conditional multinomial test of §5.3 than on the pooled estimate: dependence distorts not the mean but the null distribution of the statistic; the same truncation check covers that test as well.

4 Two observation schemes

4.1 The distinction

The pivot (10) was derived on the assumption that N_b is fixed by the observation plan: we watch the process until N_b events have accumulated, then stop. Call this *scheme A* (inverse sampling, Type-II censoring).

The data arose differently. For each base the computational search covered the whole interval $(t_0, \ln L_b]$, and every event falling inside it was found; the number of finds N_b is a random variable, while the frontier L_b is fixed. Call this *scheme B*.

Remark 5 (What here is classical and what is not). The distinction between the schemes is itself classical and goes back to reliability theory, where scheme A is called a *failure-truncated* and scheme B a *time-truncated* life test [13, 14]. The estimator $(N-1)/D$, unbiased under scheme A (Corollary 1), is a textbook fact there, and we lay no claim to it. The distribution of the MLE of the mean under Type-I censoring was studied by Bartholomew [15]. We also note an elementary fact: under scheme B with a known window the sufficient statistic is $N \sim \text{Poisson}(\lambda W)$, and $\mathbb{E}f(N) = \sum_n f(n) e^{-\mu} \mu^n / n!$ is an entire function of $\mu = \lambda W$, so it cannot coincide with W/μ , which has a pole at zero. Hence no unbiased estimator of the *mean* $1/\lambda$ exists under scheme B

(for time-censored samples from the exponential distribution see [16]); the same argument applies conditionally on D_b . Proposition 3 does not contradict this: it gives exact unbiasedness of an estimator of the *intensity*.

What we regard as new is something else. In the classical setting the moment at which the test is stopped is *known*: the time T is set by the experimenter. Here the frontier L_b is documented only for $b = 2$ and $b = 10$ (§4.5), so the full exposure cannot be read off a protocol and must instead be reconstructed—via the memorylessness of the process—from the finds themselves. That is where a correction *per base*, rather than one for the sample as a whole, comes from: there are as many unknown frontiers as there are bases. A survey of the biases generated by the edge of the observation window, though in a nonparametric setting and for the shape of the interevent distribution rather than for the intensity, is given in [17].

Dual problems have long been known. In paleontology the unknown end of a stratigraphic range is estimated from fossil finds, and the observed extreme horizon is extended by roughly one mean gap between finds [18, 19]. In ecology the unknown extinction time of a species is estimated from its sighting record [20]. The general problem of estimating a truncation point is treated in [21]. There, however, the unknown end of the window is the parameter of interest. Here it is a nuisance parameter, and it is the intensity that is estimated. Closest in technique is [22]: there the observation window starts at the first event, and the likelihood factorizes into a Poisson factor and a factor that does not depend on the intensity. We have found no work in which Type-I censoring with an unrecorded censoring time is treated as a problem of pooled estimation of an intensity across many strata.

The distinction is not cosmetic. Under scheme B, conditionally on $N_b = n$, the positions of the events are n independent uniform variables on $(t_0, \ln L_b]$, so the conditional distribution of D_b depends on L_b and *not* on λ_b : with N_b fixed, D_b is informative about the frontier, not about the intensity.

More precisely, under scheme B with a known window width $W_b = \ln L_b - t_0$ the likelihood for base b is $\lambda_b^{N_b} e^{-\lambda_b W_b}$, whereas with an unknown width it depends on the nuisance parameter $W_b \geq D_b$ and decreases monotonically in it. The joint MLE therefore sets $\hat{W}_b = D_b$ —the frontier exactly at the last find—and expression (8) turns out to be the *profile* likelihood over the nuisance parameters $W_1, \dots, W_B$, with M/S the profile MLE. The number of nuisance parameters grows with the number of bases, and the profile MLE is inconsistent: as $B \rightarrow \infty$ with $\lambda_b W_b$ bounded, it does not converge to κ . This is the classical Neyman–Scott situation [9] (for a survey see [23]), and the standard way out is to condition on a statistic sufficient for the nuisance parameters [24, 10]. This is what §4.2 implements.

Note also that §7 of the present work, where the frontier for $b = 2$ is known and used explicitly, rests on scheme B. One cannot apply both schemes in a single paper; below we adopt scheme B as matching the way the data were obtained, and construct exact inference for it.

4.2 The exact conditional law under scheme B

The key observation is that under scheme B one should condition not on the number of events but on the position of the last of them. The unknown frontier then disappears from the problem entirely. One assumption is required.

(A1) *The position of the frontier L_b was not chosen in the light of the search results: the computational budget, not the finds, determines how far in n the sieving has been carried. Formally: the window width $W_b = \ln L_b - t_0$ is a random variable with an arbitrary unknown distribution π_b (in particular, a constant) that is independent of the process of finds for base b .*

In the language of survival analysis, (A1) is the requirement that the censoring be *non-informative*: the stopping time does not depend on what the process has produced. The distributions π_b may be arbitrary and may differ between bases; they are never estimated.

Proposition 3 (Exact conditional likelihood). *Suppose that under scheme B a homogeneous Poisson process of intensity λ_b is observed on $(t_0, t_0 + W_b]$, that (A1) holds and $N_b \geq 1$, and let $D_b = t_{N_b} - t_0$ be the distance from the origin to the last event found.*

(i) *The joint density of (N_b, D_b) at (n, d) , $n \geq 1$, $d > 0$, factorizes:*

$$p(n, d) = \underbrace{e^{-\lambda_b d} \frac{(\lambda_b d)^{n-1}}{(n-1)!}}_{\text{depends on } \lambda_b} \cdot \underbrace{\lambda_b \int_{[d, \infty)} e^{-\lambda_b(w-d)} \pi_b(dw)}_{\text{density of } D_b}, \quad (11)$$

so that for fixed λ_b the statistic D_b is sufficient for W_b (and for the whole distribution π_b).

(ii) *Consequently,*

$$K_b := N_b - 1 \mid D_b \sim \text{Poisson}(\lambda_b D_b), \quad (12)$$

and this distribution depends neither on W_b nor on π_b .

(iii) *For processes that are independent across bases,*

$$K := M - B \mid (D_b)_b \sim \text{Poisson}(\kappa S), \quad S = \sum_b \frac{D_b}{\ln b}, \quad (13)$$

and $\hat{\kappa}_B = (M - B)/S$ is the conditional maximum-likelihood estimator, exactly conditionally unbiased: $\mathbb{E}[\hat{\kappa}_B \mid (D_b)_b] = \kappa$.

Proof. (i) First fix $W_b = w$. For $0 < d < d + \Delta \leq w$ and $\Delta \rightarrow 0$, the event $\{N_b = n, D_b \in [d, d + \Delta)\}$ means: exactly $n - 1$ events in $(t_0, t_0 + d)$, one in $[t_0 + d, t_0 + d + \Delta)$ and none in $[t_0 + d + \Delta, t_0 + w]$. By the independence of the increments of the Poisson process its probability equals $e^{-\lambda_b d} \frac{(\lambda_b d)^{n-1}}{(n-1)!} \cdot \lambda_b \Delta \cdot e^{-\lambda_b(w-d)} + o(\Delta)$, and for $d > w$ it is zero. By (A1), averaging over $w \sim \pi_b$ affects only the last factor and gives (11). Summation over $n \geq 1$ shows that the second factor in (11) is the marginal density of D_b . (ii) Divide (11) by the marginal density of D_b . (iii) A sum of independent Poisson variables is Poisson; the conditional log-likelihood $\ell_B(\kappa) = \sum_b [K_b \ln \kappa - \kappa D_b / \ln b] + \text{const}$ is maximized at $\kappa = K/S$, and conditional unbiasedness is immediate from $\mathbb{E}[K \mid D] = \kappa S$. $\square$

Remark 6 (Conditional and integrated likelihood). The same result is obtained by integrating the likelihood over the nuisance parameter. For a fixed window the joint density is proportional to $\lambda_b^n d^{n-1} e^{-\lambda_b W_b} \mathbf{1}_{\{W_b \geq d\}}$, and $\int_d^\infty \lambda_b^n e^{-\lambda_b W_b} dW_b = \lambda_b^{n-1} e^{-\lambda_b d}$, which is exactly the kernel of (12). This is the integrated likelihood in the sense of [25] with an improper uniform measure on W_b . Such a procedure depends, in itself, on the choice of measure: a measure uniform in $\mu_b = \lambda_b W_b$, that is, $d\mu_b = \lambda_b dW_b$, would give $\lambda_b^n e^{-\lambda_b d}$ and bring back the inconsistent estimator M/S . Here, however, the choice is not arbitrary. First, the result coincides with the *conditional* likelihood (12), which is defined without any measure and is invariant under reparametrization of W_b . Second, W_b enters (11) as a location parameter of the quantity $W_b - D_b$, and dW_b is the invariant measure of the translation group. Third, for $\pi_b(dw) = dw$ the marginal factor in (11) equals $\lambda_b \int_d^\infty e^{-\lambda_b(w-d)} dw = 1$ and does not depend on λ_b . Hence, when the frontier is unknown, the factor discarded by conditioning carries no information about κ that could be used without specifying π_b .

Remark 7 (Connection with the Neyman–Scott problem). The correction “one event per base” is of the same nature as the loss of one degree of freedom per group in the classical Neyman–Scott example [9]. For $y_{ij} \sim N(\mu_i, \sigma^2)$ the profile MLE $\hat{\sigma}^2$ is inconsistent, whereas conditioning on $\bar{y}_i$, which is sufficient for μ_i at fixed σ^2 , gives $\sum_j (y_{ij} - \bar{y}_i)^2 \sim \sigma^2 \chi_{n_i-1}^2$. In both cases the nuisance parameter of a group is one-dimensional and is a location parameter, and the statistic sufficient for it “absorbs” exactly one observation: the mean absorbs one degree of freedom, the

last event absorbs one event. The general mechanism is the conditional likelihood given a statistic that is sufficient for the nuisance parameter at a fixed value of the parameter of interest [24]. Its consistency as the number of strata grows was established under regularity conditions by Andersen [10], and the optimality of the conditional estimator in models that mix over the nuisance parameter by Lindsay [26].

Remark 8 (Inhomogeneous intensity). Argument (i) does not use homogeneity. If the intensity $\lambda_b(t; \theta)$ depends on a parameter θ , then conditionally on D_b the events in $(t_0, t_0 + D_b)$ form a Poisson process of the same intensity, and the conditional log-likelihood is

$$\ell_B(\theta) = \sum_b \left[\sum_{i=1}^{N_b-1} \ln \lambda_b(t_i; \theta) - \int_{t_0}^{t_0+D_b} \lambda_b(t; \theta) dt \right] : \quad (14)$$

the last event is excluded from the sum, and the integral stops at it. It is in this form that the variable-intensity models of §8 should be estimated.

The practical consequences of Proposition 3 deserve to be listed separately, since they remove almost all the caveats of the previous version of this work.

1. *Exact intervals without simulation.* Inverting the Poisson rate gives the Garwood interval [27]

$$\left[\frac{1}{2S} \chi_{\alpha/2, 2K}^2, \frac{1}{2S} \chi_{1-\alpha/2, 2(K+1)}^2 \right], \quad (15)$$

whose coverage is never below nominal. For our S it equals 0.9540 at $\kappa = e^\gamma$ and lies within 0.950–0.958 for $\kappa \in [1.5, 2.2]$ (the excess is the usual effect of discreteness).

2. *No remainder.* No exponentially small terms, no caveats about bases without finds: conditioning on $N_b \geq 1$ holds by construction, since D_b is defined only then. A base without a single find does not enter the conditional likelihood.
3. *No dependence on the position of the frontier.* The rule by which L_b was chosen does not enter (12) at all—provided (A1) holds.
4. *The cost of conditioning is small.* $\text{SD}(\hat{\kappa}_B) = \sqrt{\kappa/S} = 0.1270$ against $\sqrt{\kappa/S_{\text{full}}} = 0.1224$ with known frontiers standing at their expected positions, $S_{\text{full}} \approx S + B/\kappa$ (Proposition 4). Forgoing knowledge of L_b increases the standard deviation by 3.7%, or costs 7.1% of the Fisher information: B/κ out of $S + B/\kappa$, that is, roughly B events out of M . Knowledge of the frontiers is needed not so much for precision as for checking (A1) (§4.6).

If (A1) is violated, conditioning on D_b ceases to be harmless. If the search stops exactly at a find, then D_b is itself the stopping time, $\delta_b = 0$, and we are back at scheme A. If instead the search stops after a sufficiently long stretch without finds, then the stopping itself carries information about the process before D_b , and law (12) no longer holds (Remark 9).

4.3 What ignorance of the frontier costs

Besides Remark 7, the size of the correction is explained by a direct moment calculation. We retain it as a substantive interpretation and as the basis for the analysis of violations of (A1) in Remark 9.

Proposition 4. *Suppose that under scheme B the search for base b covered $(t_0, \ln L_b]$, that the window width $W_b = \ln L_b - t_0$ is fixed (under (A1), conditionally on W_b), and let $\delta_b = \ln L_b - t_{N_b}$ be the distance from the frontier to the last event found (with $\delta_b = W_b$ when $N_b = 0$). Then*

$$\Pr(\delta_b > x) = e^{-\lambda_b x} \quad (0 \leq x < W_b), \quad \Pr(\delta_b = W_b) = e^{-\lambda_b W_b}, \quad (16)$$

that is, δ_b is an exponential variable with parameter λ_b truncated at W_b , and

$$\mathbb{E}[\delta_b] = \frac{1 - e^{-\lambda_b W_b}}{\lambda_b} \leq \frac{1}{\lambda_b}. \quad (17)$$

The full exposure is $S_{\text{full}} = S + \sum_b \delta_b / \ln b = \sum_b W_b / \ln b$, with $\mathbb{E}[M] = \kappa S_{\text{full}}$ identically, and

$$\mathbb{E}\left[\kappa \sum_b \delta_b / \ln b\right] = B - \sum_b e^{-\lambda_b W_b}. \quad (18)$$

Proof. For $x < W_b$ the event $\{\delta_b > x\}$ means that the interval $(\ln L_b - x, \ln L_b]$ of length x contains no event of the process, which has probability $e^{-\lambda_b x}$. As $x \rightarrow W_b^-$ what remains is the case “no events at all”, which gives the atom in (16). Integrating the tail, $\mathbb{E}[\delta_b] = \int_0^{W_b} e^{-\lambda_b x} dx = (1 - e^{-\lambda_b W_b})/\lambda_b$, whence (17); the inequality is obvious. Next, S_{full} is not random: by construction $D_b + \delta_b = W_b$, so $\kappa S_{\text{full}} = \sum_b \lambda_b W_b = \mathbb{E}[M]$ —this is exactly the moment equation. Finally $\kappa \mathbb{E}[\delta_b] / \ln b = \lambda_b \mathbb{E}[\delta_b] = 1 - e^{-\lambda_b W_b}$, and summation over b gives (18). $\square$

In other words, ignorance of the frontier costs information equivalent to one event *per base*—whereas formula (9) subtracts one event *in total*. With $B = 15$ the difference is fourteen events, that is, 6.9%:

$$\frac{M - 1}{S} = 1.9648 \quad \text{versus} \quad \frac{M - B}{S} = 1.8381. \quad (19)$$

The remainder $\sum_b e^{-\lambda_b W_b}$ dropped in (18) is the expected number of bases for which the search produced no find at all; in our data $\lambda_b W_b \approx N_b \geq 7$, so it does not exceed $3 \cdot 10^{-3}$ against $B = 15$. We stress that in the exact formulation (12) this remainder does not arise at all: the moment calculation is needed here only for interpretation.

Remark 9 (What remains without (A1)). The previous version claimed that the family $(M - Bm)/S$ with $m \in [0, 1]$ yields a provable two-sided bracket $[(M - B)/S, (M - 1)/S]$. This claim is *false*, and we withdraw it. Inequality (17) was derived for a fixed window (or one independent of the process), that is, precisely under (A1), and then no bracket is needed: the estimator $(M - B)/S$ is correct. If (A1) is violated, $\lambda_b \mathbb{E}[\delta_b]$ can be any non-negative number.

Only the *upper* bound survives rigorously. If the frontier is a stopping time with respect to the search history (the decision to stop does not look ahead), then by Wald’s identity for the Poisson process $\mathbb{E}[N_b] = \lambda_b \mathbb{E}[W_b]$ under any stopping rule with $\mathbb{E}[W_b] < \infty$. Since $W_b = D_b + \delta_b \geq D_b$, we get $\mathbb{E}[M] \geq \kappa \mathbb{E}[S]$: in the sense of moments M/S bounds κ from above. There is no assumption-free lower bound. The rule “stop the search after a stretch of g mean intervals without finds” gives $\lambda_b \delta_b \geq g$. Moreover, waiting for such a stretch itself packs the events before D_b more densely, so that law (12) no longer holds. Table 3 shows the consequences: at $g = 2$ the estimator $(M - B)/S$ is inflated by 6.7%, and the exact test of Proposition 3 rejects a true hypothesis in 18% of cases instead of 5%. The rule “stop at the next find” ($\delta_b = 0$) biases $(M - B)/S$ downwards by 6.4%, and then the estimator $(M - 1)/S$ is the correct one.

The precise statement is therefore as follows. Under (A1) the main result (20) is exact. Under stopping “at a find” the estimator (21) is correct, and it also bounds κ from above in the sense of moments under any rule. Under stopping “after a drought” the true κ is *below* $(M - B)/S$, that is, even closer to e^γ or below it. None of the violations of (A1) creates an excess over LPW that the data would require explaining. But without documented frontiers the size of the uncertainty is not bounded from below: this is a stronger argument for publishing L_b than the one given previously.

4.4 Calibration of the old pivot

How bad the Gamma pivot (10) is when the data are generated by scheme B depends on how far the frontier has advanced—and this should be reported as a range rather than as a single

Table 3. Scheme B under various stopping rules: relative mean of the two pooled estimators, the ratio $\mathbb{E}[M]/(\kappa\mathbb{E}[S])$ (at least 1 under every rule by Wald’s identity), and the type I error of the exact two-sided Poisson test (13) at the 5% level. Budget $W_b^0 = N_b/\lambda_b$; “drought g ”: after the budget is spent the search continues until g/λ_b passes without a find. 20 000 replications per row, accuracy about $\pm 0.05\%$ for the means and ± 0.003 for the level (script `stopping_rules.py`, tag [S13], seed 20260924).

Stopping rule	$\mathbb{E}[(M-1)/S]/\kappa$	$\mathbb{E}[(M-B)/S]/\kappa$	$\frac{\mathbb{E}[M]}{\kappa\mathbb{E}[S]}$	type I error
(A1): frontier = W_b^0	1.0688	0.9999	1.0737	0.046
(A1): frontier = UW_b^0 , $U \sim \text{U}(0.5, 2)$	1.0551	1.0000	1.0582	0.043
stop at a find	0.9963	0.9362	1.0003	0.165
drought $g = 0.5$	1.0762	1.0074	1.0811	0.050
drought $g = 1$	1.0923	1.0250	1.0973	0.063
drought $g = 2$	1.1227	1.0667	1.1277	0.178

Table 4. Scheme B under seven frontier rules: type I error of the nominal 5% test built on the Gamma pivot (10), and the relative bias of the two pooled estimators. The rules are stated through the window width $W_b = \ln L_b - t_0$: “ $\mathbb{E}[\#]$ = observed” is $W_b = N_b/\lambda_b$; “+ s mean intervals” is $W_b = (N_b + s)/\lambda_b$; “ c times further in t ” is $W_b = cD_b$ with the observed D_b . 50 000 replications per row, accuracy about ± 0.003 for the level and $\pm 0.1\%$ for the means (tag [S10]).

Rule for the frontier	type I error	$\mathbb{E}[(M-1)/S]/\kappa$	$\mathbb{E}[(M-B)/S]/\kappa$
$\mathbb{E}[\#\text{events}] = \text{observed}$	0.1634	1.0691	1.0001
frontier +0.5 mean intervals	0.1620	1.0667	1.0002
frontier +1 mean interval	0.1596	1.0640	0.9998
frontier +2 mean intervals	0.1527	1.0599	0.9998
frontier +5 mean intervals	0.1359	1.0506	1.0002
frontier 1.5 times further in t	0.1342	1.0499	0.9999
frontier twice as far in t	0.1136	1.0371	1.0001
A (control)	0.0492	—	—

number. Table 4 gives both quantities at once: the type I error of the old test and the relative bias of both pooled estimators.

Three conclusions follow from Table 4. First, under scheme B a nominal 5% test rejects a true hypothesis in 11–16% of cases: the claimed precision of the intervals is overstated two- to threefold, and the more so the closer the frontier is to the last find. Second, the estimator $(M-1)/S$ is biased upwards by 4–7% under all rules, whereas $(M-B)/S$ is unbiased up to Monte Carlo noise under all rules—as Proposition 3 predicts, since conditional unbiasedness implies unconditional unbiasedness. Third, the figure “16.2%” quoted in an earlier version is the value for one particular scenario; the correct thing is to report the range. All seven rules of Table 4 place the frontier without regard to the finds, that is, they satisfy (A1); rules that violate (A1) are analysed in Remark 9 and Table 3.

4.5 Documented frontiers: an empirical check of (A1)

A dated and attributed frontier above the last find could be found for four bases (checked on 24 September 2026):

- $b = 2$: the GIMPS report at the time the data were read, $X_{\text{ver}} = 81\,648\,221$ and $X_{\text{ft}} = 141\,308\,443$ (§2);
- $b = 10$: “Search limit is 10800000, currently” (S. Batalov, A004023, 1 July 2021);

- $b = 22$ and $b = 26$: “No other terms < 100000 ” (R. Price, A127997, A127999, 2012; the last finds, 27 823 and 26 717, were made in 2007–2008).

The other mentions of search limits are either outdated and lie far below $n_{\max}$, or coincide with the last find and mark it rather than the search.

Under (A1) the quantities $\lambda_b \delta_b$ are independent and distributed as $\text{Exp}(1)$ (the truncation at W_b is negligible here), so their sum over k bases has the $\text{Gamma}(k, 1)$ distribution. Small values point to stopping “at a find” ($m < 1$), large ones to stopping “after a drought” ($m > 1$, Remark 9). The observed values are 0.143 ($b = 2$ with X_{ver} ; 0.093 with X_{ft}), 0.215 ($b = 10$), 0.737 ($b = 22$) and 0.722 ($b = 26$). Their mean is 0.44–0.45 against an expected value of one. The probability of so small a sum under (A1) is 0.10–0.11 (tag [S18]).

The conclusions must be cautious: there are four observations, the $b = 2$ frontier moves continuously, and two of the bases enter only the extended set. Nevertheless this is the first check of (A1) against data, and its result is informative about the *direction*. The documented frontiers lie not farther from but closer to the last finds than (A1) predicts. The violation that is dangerous for the main conclusion—stopping “after a drought”, which biases $(M - B)/S$ upwards—is not supported by the data: the upper tail is 0.89–0.90. If (A1) is violated at all, it is in the direction $m < 1$, and the true value lies between $(M - B)/S$ and $(M - 1)/S$.

The documented frontiers can also be used directly. For bases with a known frontier the full likelihood applies: $N_b \sim \text{Poisson}(\lambda_b W_b)$ with all events and the full window. For the others the conditional law (12) applies. The sum is again exactly Poisson. The hybrid estimate with the known frontiers for $b = 2$ (X_{ft}) and $b = 10$ is $\hat{\kappa} = 1.853$ with an exact 95% interval [1.608, 2.125] and $p = 0.588$ against e^γ . It differs from $(M - B)/S = 1.838$ by 0.8%, and the main conclusion does not change.

4.6 The data do not choose the scheme for us

We stress that $(M - 1)/S$ and $(M - B)/S$ are functions of the same sufficient statistics (N_b, D_b) . No processing of the available data will say which of them is right: the choice is determined by how the data were collected, and its price is 6.9%—twice the entire residual gap with LPW (3.2%).

The natural attempt to sidestep the choice is to build the pooled estimate out of pieces that are unbiased by construction. The per-base estimator $\tilde{\kappa}_b = (N_b - 1) \ln b / D_b$ is unbiased under scheme A (Corollary 1) and unbiased under scheme B up to an exponentially small term. Indeed, for $N_b = n \geq 2$ the ratio D_b / W_b is distributed as $\text{Beta}(n, 1)$, and the identity $\mathbb{E}[1/\text{Beta}(\alpha, \beta)] = (\alpha + \beta - 1)/(\alpha - 1)$ gives $\mathbb{E}[\tilde{\lambda}_b \mid N_b = n] = n / W_b$; for $n \in \{0, 1\}$ the estimator is zero. Averaging over $n \sim \text{Pois}(\lambda_b W_b)$ therefore gives not λ_b but

$$\mathbb{E}[\tilde{\lambda}_b] = \lambda_b - \frac{\Pr(N_b = 1)}{W_b} = \lambda_b (1 - e^{-\lambda_b W_b}),$$

that is, the same exponentially small deficit: for $\lambda_b W_b \approx N_b \geq 7$ it does not exceed 0.1%.

Nevertheless, the weighted mean

$$\kappa_* = \sum_b (N_b - 2) \tilde{\kappa}_b / \sum_b (N_b - 2)$$

with inverse-variance weights has a bias of 0.0% under scheme A but +7.5% under scheme B, that is, practically the same as $(M - 1)/S$. The reason is that the weights—the event counts themselves—are random under scheme B and positively correlated with the estimates; unbiasedness of each summand does not carry over to a ratio of sums. The correct way to pool is not to weight the per-base estimates but to sum the sufficient statistics, as (13) does.

From this we draw a narrow but, we think, important conclusion: *the gain in precision from pooling the bases is inseparable from an assumption about the observation scheme*. The only radical remedy is to document the search frontiers L_b , as GIMPS does for $b = 2$. We address this request to the participants of the relevant search projects.

Table 5. Per-base estimates. $\hat{G}_b = D_b/(N_b \ln b)$ is the endpoint form; $\hat{G}_b^{\text{OLS}}$ is the OLS slope of $\ln n_k$ on k divided by $\ln b$; both are unbiased (Proposition 2). $\hat{\kappa}_b = K_b/(D_b/\ln b)$ is the scheme B conditional estimator with the exact Garwood interval (15) from $\text{Poisson}(\kappa D_b/\ln b)$. The last column is the Poisson mid-PIT at $\kappa = e^\gamma$: values near one correspond to bases with a *high* κ_b (orientation opposite to the Gamma PIT of the earlier version). LPW predicts $G = e^{-\gamma} = 0.5615$ and $\kappa = e^\gamma = 1.7811$.

b	N_b	K_b	$\hat{G}_b$	$\hat{G}_b^{\text{OLS}}$	diff., %	$\hat{\kappa}_b$	95% CI (Garwood)	mid-PIT
2	51	50	0.5102	0.5309	+4.1	1.921	[1.426, 2.533]	0.710
3	22	21	0.6148	0.6081	−1.1	1.552	[0.961, 2.373]	0.272
5	18	17	0.4802	0.4666	−2.8	1.967	[1.146, 3.149]	0.670
6	16	15	0.5000	0.4904	−1.9	1.875	[1.049, 3.092]	0.594
7	9	8	0.7104	0.7371	+3.8	1.251	[0.540, 2.466]	0.160
10	10	9	0.6612	0.6746	+2.0	1.361	[0.622, 2.584]	0.216
11	12	11	0.4034	0.3617	−10.3	2.272	[1.134, 4.066]	0.794
12	13	12	0.3981	0.3796	−4.7	2.319	[1.198, 4.050]	0.820
13	12	11	0.4098	0.3448	−15.9	2.237	[1.117, 4.002]	0.780
14	10	9	0.5025	0.5386	+7.2	1.791	[0.819, 3.400]	0.528
15	9	8	0.5035	0.5054	+0.4	1.766	[0.762, 3.479]	0.513
17	11	10	0.4301	0.4353	+1.2	2.114	[1.014, 3.887]	0.717
18	7	6	0.6604	0.4656	−29.5	1.298	[0.476, 2.825]	0.228
19	10	9	0.3161	0.2971	−6.0	2.847	[1.302, 5.405]	0.911
20	8	7	0.5300	0.5713	+7.8	1.651	[0.664, 3.402]	0.444

5 Results

5.1 Per-base estimates

Eleven of the fifteen bases lie below $e^{-\gamma}$ under the endpoint form, and eleven of the fifteen under OLS—but these are *different* elevens. The membership changes: base 18 lies above $e^{-\gamma}$ under the endpoint form and below it under OLS, and base 20 the other way round. In other words, even the sign of the deviation is not fully robust to the choice of form; only their *number* is. This is the same effect as in §3.4, and it affects exactly the two bases for which the choice of form weighs more than the deviation itself.

All fifteen mid-PIT quantiles lie inside the central 95% region; the extremes are $b = 19$ at quantile 0.911 and $b = 7$ at 0.160. Base 13, the “lucky” one, sits at 0.780, which is in no way remarkable for one base out of fifteen. All fifteen Garwood intervals cover e^γ .

5.2 The pooled estimate

The exact conditional inference for scheme B (Proposition 3) gives

$$\hat{\kappa}_B = \frac{M - B}{S} = 1.8381, \quad 95\% \text{ CI} = [1.5939, 2.1091], \quad p = 0.672, \quad (20)$$

where the interval is (15) with $K = 203$, $S = 110.4423$, and the p -value is obtained by the doubled-minimum-tail convention for $K \sim \text{Poisson}(e^\gamma S)$, $e^\gamma S = 196.71$; under the mid- p convention $p = 0.647$. The exact coverage of interval (20) is 0.9540 at $\kappa = e^\gamma$ and does not fall below 0.950 for any κ .

For comparison, scheme A via (9)–(10) gives

$$\tilde{\kappa} = 1.9648, \quad 95\% \text{ CI} = [1.7205, 2.2444], \quad p = 0.142. \quad (21)$$

The point estimate (21) is 10.3% above e^γ , and it is precisely this excess that used to be interpreted as the trace of a factor omitted by the heuristic. On moving to scheme B the excess falls to 3.2%, and the p -value rises from 0.14 to 0.67.

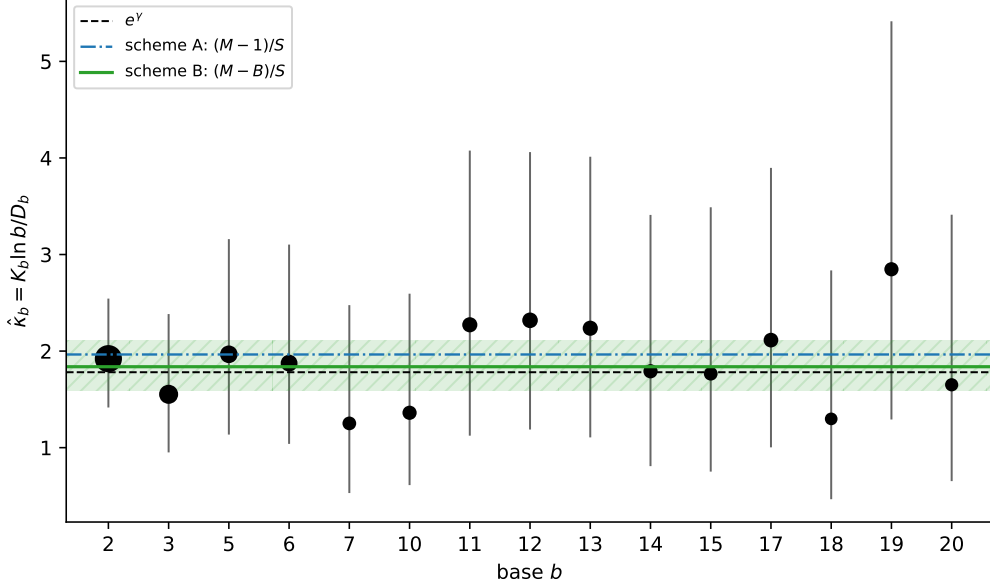


Figure 1. Per-base estimates $\hat{\kappa}_b$ with exact 95% Garwood intervals (15); marker area is proportional to N_b . Dashed line: the LPW value e^γ ; green with a band: the pooled scheme B estimate and its interval; blue dash-dotted: the scheme A estimate $(M-1)/S$. Each base separately is compatible with a universal constant.

We hold that scheme B is closer to the way the data were obtained, and therefore regard (20) as the main result. Estimate (21) corresponds to stopping the search exactly at a find and, by Remark 9, bounds κ from above in the sense of moments under any stopping rule; under stopping “after a drought” the true value lies below (20). Without publication of the L_b , (A1) cannot be checked. Both estimates are consistent with LPW; they differ in whether the data call for an explanation of a residual excess. We conclude that they do not. The overall picture is shown in Figure 1: the confidence interval of every base covers both e^γ and both pooled estimates.

5.3 Homogeneity, and its power

Let H_0 be the model with a single κ and H_1 the model with free κ_b . From the conditional likelihood of Proposition 3, since $\sum_b \hat{\kappa}_b D_b / \ln b = K = \hat{\kappa}_B S$, the likelihood-ratio statistic reduces to

$$\Lambda = 2 \sum_b K_b \ln \frac{\hat{\kappa}_b}{\hat{\kappa}_B} = 2 \sum_b K_b \ln \frac{K_b}{K s_b / S}, \quad s_b = \frac{D_b}{\ln b}. \quad (22)$$

Crucially, its null distribution is *exact* here: conditioning on $K = \sum_b K_b$ turns (13) into the multinomial model $(K_b) \mid K \sim \text{Multinomial}(K; s_b/S)$, and there is no need to appeal to χ^2_{B-1} asymptotics. We obtain $\Lambda = 7.127$, exact $p = 0.935$ ($2 \cdot 10^6$ multinomial replications, ± 0.0002); the asymptotic approximation gives $p = 0.930$ on 14 degrees of freedom, that is, it works well in this case.

But a non-rejection is informative exactly to the extent that the test is powerful, and here one must apply to oneself the same requirement one applies to the literature. We simulated alternatives in which the κ_b are lognormally scattered about e^γ with coefficient of variation cv :

cv	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.50
power	0.059	0.097	0.184	0.326	0.484	0.645	0.774	0.851	0.948

The test attains 80% power only at $cv \approx 0.37$ (12000 replications per grid point, power accurate to about ± 0.004). Hence $p = 0.935$ means: the data do not require base-dependent

Table 6. Robustness to lower truncation. A base is retained if at least *two* events survive the truncation ($N_b \geq 2$); at $n > 10^4$ base 19 no longer meets this criterion, whence $B = 14$. The criterion matters: under the weaker requirement $N_b \geq 1$ base 19 is retained, giving the last row. The interval and p_B are exact, scheme B; the last column (p_A , the scheme A Gamma pivot) is given for comparison with the old practice.

truncation	B	M	$(M - 1)/S$	$(M - B)/S$	95% CI, scheme B	p_B	p_A
none	15	218	1.9648	1.8381	[1.594, 2.109]	0.672	0.142
$n > 10$	15	192	2.0192	1.8712	[1.606, 2.168]	0.531	0.080
$n > 10^2$	15	156	2.0116	1.8299	[1.540, 2.158]	0.771	0.124
$n > 10^3$	15	125	2.0551	1.8231	[1.498, 2.197]	0.832	0.106
$n > 10^4$	14	89	2.0990	1.7889	[1.407, 2.242]	1.000	0.116
$n > 10^4, N_b \geq 1$	15	90	2.0848	1.7569	[1.382, 2.202]	0.966	0.128

frequencies, but neither do they exclude a spread of κ_b exceeding a third of the value of the constant. The statement “lucky bases do not exist” does not follow from the available data; what follows is the weaker statement “their existence is not supported by the data, and differences below $\pm 37\%$ are in principle indistinguishable”.

5.4 Robustness

Lower truncation. We varied the lower truncation point in order to probe the asymptotic regime (the heuristic (2) is asymptotic, whereas the constant-intensity model starts at $n = 2$). The results are in Table 6. The scheme A estimate grows from 1.965 to 2.099—not strictly monotonically (at $n > 10^2$ it retreats from 2.019 to 2.012) but systematically: the growth is 6.8%, and all five values come from nested subsamples of the same data. This is a trend, not noise. The scheme B estimate shows no upward trend and stays close to e^γ , within 1.789–1.871 (under the weaker dropout criterion $N_b \geq 1$, within 1.757–1.871), and its p -values lie between 0.53 and 1.00 at all truncation levels. The reason is visible from (13): truncation reduces M but not B , whereas scheme A subtracts one regardless of the sample size. We regard this as confirmation of the diagnosis of §4: the upward trend was a property of the estimator, not of the data. From $n > 10$ onwards the scheme B estimate, on the contrary, *decreases* monotonically ($1.871 \rightarrow 1.789$). This is exactly what the second-order term of the heuristic itself requires, and once that term is taken into account the decrease disappears (§9). Both curves are shown in Figure 2.

Leave-one-base-out. A leave-one-base-out analysis gives $\hat{\kappa}_B$ within [1.808, 1.878] (the extreme values arise when $b = 19$ and $b = 3$ are removed) and exact p -values within [0.49, 0.85]; no single base determines the conclusion.

Status of the data. Restricting base 2 to the fifty double-checked exponents moves the result from $\tilde{\kappa} = 1.965$, $p = 0.142$ to $\tilde{\kappa} = 1.961$, $p = 0.151$ (scheme A) and leaves the scheme B estimate practically unchanged. Excluding all indices $n > 10^6$, where the entries have PRP status, gives $\tilde{\kappa} = 1.951$ with $M = 182$. In the worst case, when the last known PRP of some base turns out to be composite, the largest shift comes from $b = 17$: $\tilde{\kappa} = 1.977$. All variants lie inside interval (21), and a fortiori inside (20), so possible pseudoprimes do not affect the conclusion.

6 Statistical power

By (13) the power against a true constant $\kappa = re^\gamma$ at level α is available in closed form (Table 7). The convention has to be stated: throughout this work the test is *two-sided*, with the doubled-

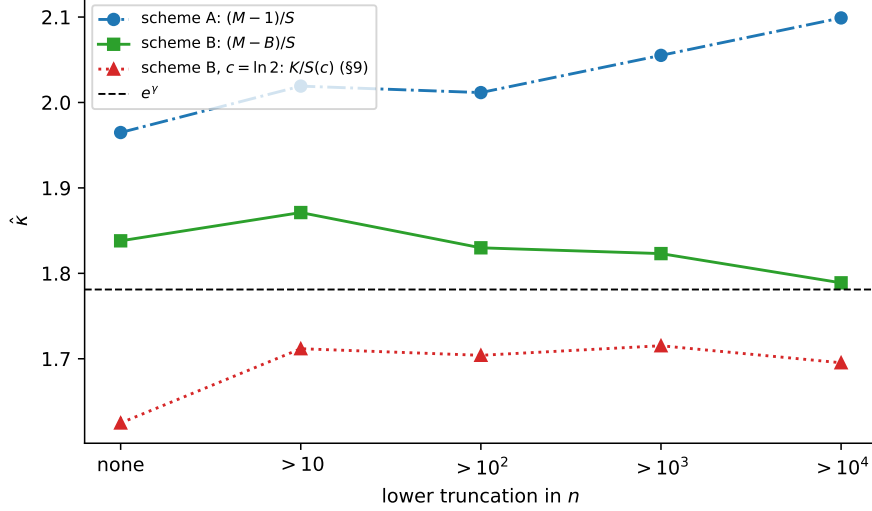


Figure 2. The pooled estimate under lower truncation: $(M-1)/S$ (scheme A), $(M-B)/S$ (scheme B) and $K/S(c)$ with the second-order term $c = \ln 2$ (§9). The upward trend of scheme A disappears under scheme B; the residual decline of scheme B disappears once the second-order term is included.

Table 7. Power at level $\alpha = 0.05$ for rejecting $\kappa = e^\gamma$ when the true constant exceeds it by the stated factor. Exposures: $S_2 = 26.02$ ($K = 50$) for base 2 alone and $S = 110.44$ ($K = 203$) for the pooled sample.

κ/e^γ	two-sided		one-sided	
	Mersenne only	pooled	Mersenne only	pooled
1.05	0.060	0.099	0.082	0.165
1.10	0.099	0.266	0.147	0.386
1.15	0.164	0.514	0.235	0.644
1.20	0.253	0.752	0.343	0.844
1.30	0.479	0.972	0.581	0.988
1.50	0.861	1.000	0.910	1.000

minimum-tail convention, $\alpha = 0.05$. The one-sided variant of the same test gives noticeably different numbers; they are shown in the right-hand part of the table for comparability with works in which the convention is not stated.

At 80% power the Mersenne sequence alone detects only deviations of 45.5% and above; pooling fifteen bases lowers that threshold to 21.3% (one-sided variant: to 18.7%). Attaining 80% power against a 10% deviation requires an exposure $S \approx 469$, that is, $K \approx 835$ conditional events ($M \approx 850$) at $\kappa = e^\gamma$ (if the true constant is 10% larger, there will be ≈ 919 events); against 5%, $S \approx 1816$ and $K \approx 3234$. The power function of a discrete test is not monotone in S (it is saw-toothed), so the values given are those beyond which the power no longer drops below 80%.

These thresholds are computationally out of reach, and it is worth saying by how much. Since K grows logarithmically in the search frontier, doubling the frontiers of *all fifteen bases* yields $e^\gamma \ln 2 \sum_b 1/\ln b = 9.64$ additional events for the entire pooled sample. To gather the missing 632 events one needs $632/9.64 \approx 65.6$ doublings, that is, a growth of the frontiers by a factor of $2^{65.6} \approx 5.5 \cdot 10^{19}$; for a sensitivity of 5%, about 314 doublings, that is, a factor of order 10^{94} . This is not a matter of decades but of magnitudes lying beyond the reach of any conceivable computation. Both power curves, as functions of the true deviation, are shown in Figure 3.

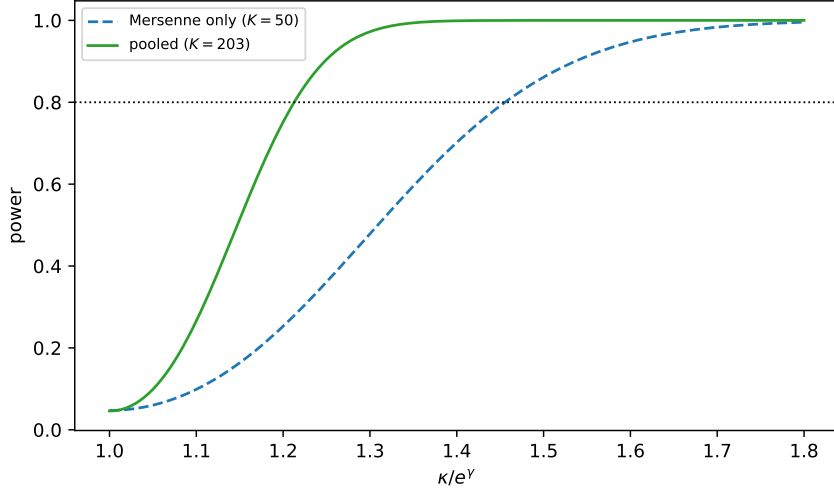


Figure 3. Power of the two-sided rejection of $\kappa = e^\gamma$ at level $\alpha = 0.05$. The horizontal line marks 80% power.

Hence a reading rule for this literature. A report that the data agree with LPW to within a few percent is not evidence for LPW at the level of a few percent; it is compatible with a constant that is wrong by a factor of one and a half. The pooled analysis does, however, establish a genuine bound: the universal constant, if it exists, lies roughly in $[1.59, 2.11]$, and no base deviates from it detectably.

7 The same error in a different guise

The Mersenne counting function invites a related error, which we present as a worked example. Regressing $\pi_2(t)$ on t at the event points, over the 50 double-checked exponents (numbered $0, \dots, N-1$: the smallest exponent is the origin, not an event), gives a slope of 2.6391 and an intercept of -2.9627 with an OLS standard error of 0.4469—apparently significant, and suggestive of a secondary term in (2). Nothing of the sort.

Two defects compound. The residuals of the cumulative count are strongly autocorrelated, here $\rho_1 = 0.825$; the Newey–West covariance estimator with eight lags raises the standard error to 0.7315. More importantly, the regression uses only the event points and thereby discards the empty interval between the last exponent and the search frontier. The correct null model holds the interval $[t_0, T]$ fixed, where $T = \ln X_{\text{ver}}$, and conditions on exactly $N-1 = 49$ events strictly to the right of t_0 , which, by the order-statistics property of the Poisson process given the number of events (cf. the proof of Proposition 2(b)), means 49 independent uniform points on $(t_0, T]$. Over 60 000 such replications the intercept has median -1.529 and a 5–95% range of $[-5.99, 2.20]$, giving a two-sided $p = 0.570$ for the observed value. The secondary term is an artifact of conditioning.

7.1 Three consequences of one and the same conditioning

(i) The bootstrap does not help. The natural reaction to a suspiciously small OLS standard error is to bootstrap over the event points. On the base-2 data, a bootstrap over 20 000 resamples gives median -2.948 , standard deviation 0.385 and a 95% interval $[-3.809, -2.300]$ lying entirely below zero; the proportion of negative values is 100%. This is not a confirmation but a repetition of the same error: resampling the observed points reproduces the variability within the realized configuration and cannot recover the variability of the process that produced that configuration. The correct null distribution has standard deviation 2.499, that is, it is 6.5 times wider than the

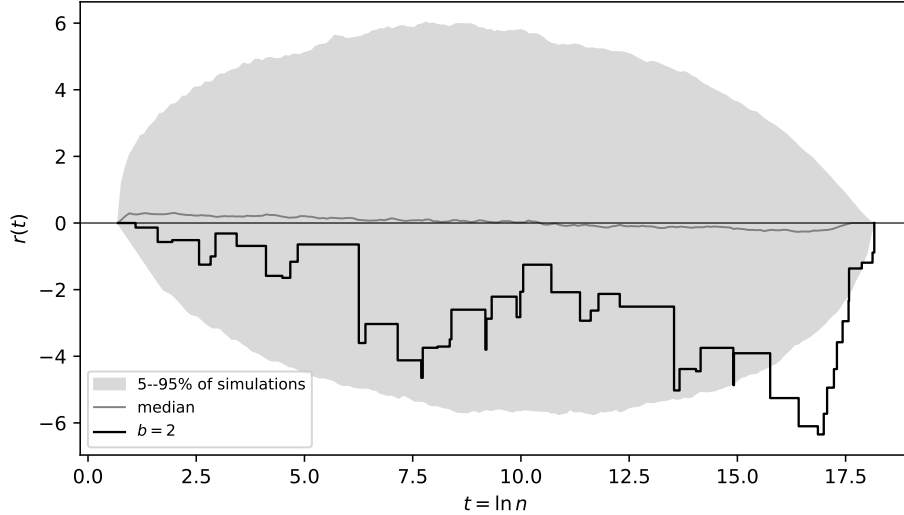


Figure 4. The residual process $r(t)$ for base $b = 2$ together with the 5–95% quantile band of the simulations and the simulation median. The trajectory stays inside the band.

bootstrap one *in terms of standard deviation* (and likewise 6.5 times in terms of the width of the 95% interval).

(ii) Spurious underdispersion. Comparing the observed $\max_i |r_i| = 6.347$, where $r_i = i - \hat{\lambda}(t_i - t_0)$, with a simulation of an *unconditioned* Poisson process suggests the conclusion that the counting function is “smoother than Poisson”. Against the correct null distribution—conditioning on $N - 1$ events and estimating λ from the last of them—the median is 5.117 and the two-sided significance level is 0.529. The observation is entirely ordinary.

(iii) The “mid-range deficit”. Estimating λ from the last event forces the residual r to vanish at both ends ($r(t_0) = r(t_N) = 0$ under the correct numbering), turning its trajectory into an analogue of a Brownian bridge, for which a negative excursion in the middle of the range arises of its own accord. The observed mean excursion over the middle half of the range is -3.13 against a simulation median of -0.012 and $p = 0.272$; the proportion of replications with a negative mean excursion is 0.502, so there is no systematic deficit. The trajectory of the residual process itself, together with the simulation band, is shown in Figure 4.

7.2 Base 2 at a known frontier

Base 2 is the only one whose frontier is published, and hence the only one for which a scheme B analysis is possible *using the full exposure* rather than only the conditional one. With $T = \ln X_{\text{ver}}$ we obtain $\hat{\lambda} = N/(T - t_0) = 2.7960$ against $e^\gamma/\ln 2 = 2.5695$, that is, $\hat{\kappa}_2 = 1.9381$ with an exact 95% Garwood interval $[1.434, 2.562]$ and a Poisson p -value of 0.593 (doubled tail; under the mid- p convention, 0.545—the convention must be stated here, since the difference is appreciable). The conditional estimate on *the same slice of data*—the same 50 double-checked exponents—gives 1.9046; the difference is small because the last exponent of that slice, 77 232 917, happens to lie very close to the frontier 81 648 221. (For comparison, the conditional estimate over all 52 known exponents, reported in Table 5, is 1.9215; it must not be compared with $\hat{\kappa}_2$, being a different sample.) This is a coincidence of one realization, not a property of the method: the correction of §4.3 concerns the average over the position of the last event relative to the frontier.

We note in passing that the regime described here is exactly the one to which we urge a transition for all bases: with W_b known, conditioning on D_b becomes unnecessary, and the

Table 8. Comparison of structural models under the conditional Poisson likelihood of Proposition 3 (full form, including $-\ln K_b!$); LR and p are relative to model A; BIC is computed with $n = K = 203$.

Model	parameters	$\ln L$	LR	p	AIC	BIC	$P(\text{model} \mid \text{data})$
A	1	-35.753	—	—	73.51	76.82	0.906
B	3	-35.162	1.18	0.554	76.32	86.26	0.008
C	2	-35.458	0.59	0.443	74.92	81.54	0.085
D	4	-34.797	1.91	0.591	77.59	90.85	0.001

precision increases by the very 3.7% discussed in §4.2.

7.3 The theoretical secondary term has the opposite sign

There is an independent argument against a substantive reading of the intercept C_1 . If the secondary term in (2) originated in the Mertens constant $\mathfrak{M} = 0.26150\dots$, its natural size would be $(e^\gamma / \ln 2)\mathfrak{M} = +0.672$: the sign is positive, whereas what is observed is -2.963 . The nearest theoretical candidates not merely fail to reproduce the observed magnitude—they predict the opposite sign. Independently of the foregoing, this confirms that C_1 is a parameter of the fit and not of the structure.

8 Structural corrections

Consider the more general parametrization of the intensity $\lambda_b = (\kappa / \ln b) C_b$ and four models for the factor:

$$\begin{aligned}
\text{A: } C_b &= 1; \\
\text{B: } C_b &= \prod_{p|b-1, p \neq 2} \left(\frac{p}{p-1}\right)^\alpha \prod_{p|b+1, p \neq 2} \left(\frac{p}{p-2}\right)^\beta; \\
\text{C: } C_b &= \prod_{p|b^2-1} \left(\frac{p}{p-1}\right)^\delta; \\
\text{D: } C_b &= \exp(\theta_1 \omega(b-1) + \theta_2 \omega(b+1) + \theta_3 \ln b).
\end{aligned}$$

The exponent in model C is written δ rather than γ so as not to confuse it with the Euler constant. The likelihood here is the *conditional Poisson* likelihood of Proposition 3, $\ell_B = \sum_b [K_b \ln \mu_b - \mu_b - \ln K_b!]$ with $\mu_b = \kappa C_b D_b / \ln b$; the previous version used the scheme A likelihood (8), which is incompatible with §4. The results are in Table 8. No parametrization gives a significant improvement; the BIC-based Bayes factors at $n = K = 203$ overwhelmingly support the base model.

Remark 10 (Sensitivity of the BIC). The choice of effective sample size for the BIC is not obvious: the likelihood has $B = 15$ independent summands, but the information grows as $K = 203$. With $n = 15$ the posterior probabilities are 0.661 / 0.080 / 0.229 / 0.030; with $n = 203$ they are 0.906 / 0.008 / 0.085 / 0.001. The conclusion favors model A under either choice, but with $n = 203$ it is substantially stronger. We report $n = K$ as matching the number of independent conditional events.

A fixed arithmetic correction. We considered separately a model with a *fixed* (not estimated) factor $C_b^{\text{fix}} = \prod_{p|b-1, p \neq 2} \frac{p}{p-1} \prod_{p|b+1, p \neq 2} \frac{p}{p-2}$, whose mean over the fifteen bases is 2.462. It gives $\hat{\kappa} = 0.759$. We stress that comparing this value with e^γ is meaningless: in the reparametrized model κ has a different scale, and such a “goodness-of-fit test” would only be

testing whether the mean of C_b equals one. Since models A and C^{fix} are non-nested and have the same number of free parameters, the correct comparison is directly by likelihood: $\ln L = -51.53$ against -35.75 , a difference of -15.78 . The fixed correction is decisively rejected.

Logarithmic corrections. A substantive correction is needed here. The model $\lambda_b(t) = (\kappa/\ln b)(1 + c_1/t + c_2/t^2)$ used in the previous version is *not an admissible intensity*: at the estimates obtained there, $\hat{c}_1 = +0.108$, $\hat{c}_2 = -0.826$, the factor vanishes at $t = 0.8564$ and is negative to the left of that point, that is, for $n < 2.355$; in particular, at $t_0 = \ln 2$ it equals -0.5634 . The bases with $n_{\min} = 2$ ($b = 2, 6, 10, 12, 18$) integrate the intensity over a region where it is negative, and the measure $\int \lambda dt$ is negative on that stretch. That the likelihood could be computed at all was a matter of chance: the smallest index that is an event is 3, that is, $t = 1.0986 > 0.8564$, so no negative values fell under the logarithm.

The correct parametrization of the same family is

$$\lambda_b(t) = \frac{\kappa}{\ln b} \exp\left(\frac{c_1}{t} + \frac{c_2}{t^2}\right), \quad (23)$$

which is positive identically and reduces to the null model at $c_1 = c_2 = 0$. It is estimated by the conditional point-process likelihood (14): the sum $\sum \ln \lambda(t_i)$ is taken over the events *excluding the last*, and the integral $\int \lambda$ runs from t_0 to the last event $t_0 + D_b$, not to the unknown frontier (Remark 8; the likelihood of the count alone is inadmissible here: it discards the positions of the events, which are the only thing that determines c_1, c_2). The result: $\hat{\kappa} = 1.748$, $\hat{c}_1 = 0.75$, $\hat{c}_2 = -1.44$; the likelihood ratio against $c_1 = c_2 = 0$ is 2.03 on two degrees of freedom, $p = 0.36$ (tag [S17]). There is no significant improvement. Note, however, that $\hat{c}_1$ is close to the value $\ln 2$ predicted by the heuristic itself; this topic is taken up in §9.

9 The second-order term of the heuristic itself

So far we have tested the constant-intensity model (3). But the LPW heuristic itself predicts an intensity that is *not* constant at finite n , and this changes the meaning of comparing $\hat{\kappa}$ with e^γ .

Derivation. Let p be prime, $p \nmid b-1$, and let q be a prime divisor of $R_b(p)$. Then b has order p modulo q , whence $q \equiv 1 \pmod{p}$, and since $R_b(p)$ is odd, $q \equiv 1 \pmod{2p}$. Hence $R_b(p)$ has no prime divisors below $2p+1$, and by Mertens' theorem the probability of primality is increased by the factor $\prod_{q < 2p} (1 - 1/q)^{-1} \approx e^\gamma \ln(2p)$ compared with a random number of the same size. This is the derivation of [1, 3]:

$$\Pr(R_b(p) \text{ prime}) \approx \frac{e^\gamma \ln(a_b p)}{p \ln b}, \quad a_b \geq 2. \quad (24)$$

Wagstaff's refinement for $b = 2$ takes into account that $q \equiv \pm 1 \pmod{8}$ and gives $a = 2$ for $p \equiv 3 \pmod{4}$ and $a = 6$ for $p \equiv 1 \pmod{4}$. For general b the analogous necessary condition is that b be a quadratic residue modulo q , since $b^p \equiv 1$ and p is odd; the smallest admissible candidate $2kp + 1$ determines $a_b = 2k$. Multiplying (24) by the density of primes $1/\ln n$ and passing to $t = \ln n$, we get

$$\lambda_b(t) = \frac{\kappa}{\ln b} \left(1 + \frac{c_b}{t}\right), \quad c_b = \langle \ln a_b \rangle, \quad \kappa = e^\gamma, \quad (25)$$

and, for the counting function, $\pi_b(x) \approx (e^\gamma/\ln b)(\ln x + c_b \ln \ln x)$. The asymptotics (2) are unaffected, but at $t \approx 10$ – 18 the correction c_b/t amounts to 4 to 17%—as much as or more than the effects discussed in this paper. Averaging over primes $p \in [10^5, 2 \cdot 10^5]$ gives c_b from 1.23 ($b = 19$) to 1.74 ($b = 15$), and for $b = 2$ the value $1.242 = \frac{1}{2}(\ln 2 + \ln 6)$, as it should be (script `lpw_second_order.py`, tag [S14]). The value $c = \ln 2 \approx 0.693$ is a lower bound that uses only $q \equiv 1 \pmod{2p}$. We stress that (24) is a heuristic; its second-order term is less reliable than the leading one, and we treat c as an external parameter in the range $[\ln 2, \langle \ln a_b \rangle]$.

Table 9. Exact conditional inference for three values of the second-order term (25): the estimate $\hat{\kappa} = K/S(c)$ from (26) and the exact two-sided p against $\kappa = e^\gamma$. “W” is Wagstaff’s $c_b = \langle \ln a_b \rangle$. Truncation as in Table 6. Tag [S16].

truncation	K	$\hat{\kappa}(c=0)$	$\hat{\kappa}(c=\ln 2)$	$\hat{\kappa}(W)$	$p(c=0)$	$p(c=\ln 2)$	$p(W)$
none	203	1.838	1.625	1.453	0.672	0.200	0.003
$n > 10$	177	1.871	1.712	1.578	0.531	0.630	0.111
$n > 10^2$	141	1.830	1.704	1.595	0.771	0.636	0.201
$n > 10^3$	110	1.823	1.715	1.620	0.832	0.738	0.344
$n > 10^4$	75	1.789	1.695	1.612	1.000	0.723	0.421

Exact conditional inference for a given c . By Remark 8, conditionally on D_b the interior events form a Poisson process of intensity (25) on $(t_0, t_0 + D_b)$. Since $\int_{t_0}^{t_0+D_b} (1 + c/t) dt = D_b + c \ln((t_0 + D_b)/t_0)$, Proposition 3 carries over verbatim:

$$K \mid (D_b)_b \sim \text{Poisson}(\kappa S(c)), \quad S(c) = \sum_b \frac{D_b + c_b \ln((t_0 + D_b)/t_0)}{\ln b}, \quad (26)$$

the estimator is $K/S(c)$, and the Garwood interval (15) applies with S replaced by $S(c)$.

Results. Table 9 yields four conclusions.

(i) *The second-order term is not determined by the data.* The profile MLE of a common c from the conditional likelihood (14) on all the data is -0.40 with a 95% interval $[-0.69, 0.44]$ (the left end is the admissibility boundary $1 + c/t_0 > 0$). But already at $n > 10$ the interval widens to $[-1.68, 2.48]$, and at $n > 10^3$ to $[-5.3, 20]$ (tag [S16]). The positions of the events within the sequences carry almost no information about c : a shift in c is compensated by a shift in κ . The two-parameter model (23) gives no significant improvement either (§8). The value of c must therefore come from theory, not from the data.

(ii) *The inference about κ depends on it more strongly than on the observation scheme.* At $n > 10$, moving from $c = 0$ to $c = \ln 2$ lowers the estimate by 8.5%, and moving to $c = \langle \ln a_b \rangle$ lowers it by 15.7%. This exceeds the price of the choice of observation scheme (6.9%, §4.3) and is two to three times the residual gap between $(M - B)/S$ and e^γ (3.2%). Comparing the constant-intensity estimate with e^γ is not a test of LPW at finite n : LPW itself predicts that such an estimate lies *above* e^γ .

(iii) *LPW with its own second-order term is consistent with the data.* For $c = \ln 2$ and $n > 10$ the exact p lies within 0.63–0.74. For $c = \langle \ln a_b \rangle$ it is 0.11–0.42. On the full sample with $c = \langle \ln a_b \rangle$ one obtains $p = 0.003$. But the full sample includes $n \leq 10$, where expansion (25) is meaningless: at $n = 2$ the factor $1 + c/t$ equals 2 even for $c = \ln 2$, and the indices 2, 3, 5, 7 themselves are dictated by the arithmetic of small numbers. We do not regard this result as evidence against LPW; it is a warning against using small indices when $c \neq 0$.

(iv) *The trend under truncation is explained by the second-order term.* For $c = 0$ the estimate decreases monotonically as the lower bound grows, from 1.871 at $n > 10$ to 1.789 at $n > 10^4$ (−4.4%). This is exactly what one expects from (25): the mean of $1/t$ over the exposure falls as t_0 grows. For $c = \ln 2$ the same four estimates lie within 1.695–1.715 (a range of 1.2%). The subsamples are nested, so this observation is illustrative rather than a separate test.

Consequence for the folklore. The term $c_b \ln \ln x$ in the counting function also explains the observation with which this work began: the normalized frequencies lie below $e^{-\gamma}$ and converge to it slowly from below. Under (25) the slope of t_k against k is $e^{-\gamma} \ln b / (1 + c_b \langle 1/t \rangle)$, so the normalized frequency is depressed by the factor $1 + c_b \langle 1/t \rangle$ and tends to $e^{-\gamma}$ as $1/t$, that is, as $1/\ln n$ —slowly indeed. In §3 we showed that fitting up to the right endpoint does not create the effect; here we see where it comes from: from the heuristic itself, not from $\kappa \neq e^\gamma$.

Summary. The question “are the data compatible with LPW?” splits into two: one about the leading term, $\kappa = e^\gamma$, and one about the second-order term, c_b . The latter is not determined by the data, but it affects the answer to the former more strongly than everything analysed in §4. For any $c \in [\ln 2, \langle \ln a_b \rangle]$ and $n > 10$, LPW is not rejected. Refining the second-order term is a problem for number theory, not for statistics. Without it, a quantitative test of LPW more precise than $\pm 10\%$ remains out of reach even as the frontiers grow: the term c_b/t decays only as $1/\ln n$.

10 Conclusions

1. The estimators of the normalized frequency of generalized repunit primes that are in common use are *not* biased by the fitting scheme: both the endpoint form and the OLS slope are exactly unbiased under scheme A and unbiased up to an error exponentially small in μ under scheme B (at most 0.73% for the endpoint form and 2.3% for OLS when $\mu \geq 7$). Only the reciprocal is biased, and that is a consequence of convexity. The common explanation of “slow convergence from below” in terms of censoring fails under both schemes.
2. The genuine defect of current practice is that different forms of the estimator disagree by up to 29.5%, and that point values are published without confidence intervals even though the OLS standard errors in this problem are understated: the Newey–West correction increases them by a factor of 1.6, and the correct conditional null distribution is 5.6 times wider than the nominal one (§7). The classification of bases into “lucky” and “unlucky” flips when the form of the estimator is changed on unchanged data; for two of the fifteen bases even the sign of the deviation from $e^{-\gamma}$ flips.
3. The main result is the exact conditional likelihood for scheme B. The unrecorded frontiers are Neyman–Scott nuisance parameters, and the profile estimator M/S is inconsistent. The position D_b of the last event is sufficient for the frontier, and conditionally on it $N_b - 1$ is distributed as $\text{Poisson}(\lambda_b D_b)$ under any rule for choosing the frontier that does not depend on the finds. The same result is obtained by integrating over the frontier against the invariant measure dW_b . Hence $M - B \mid D \sim \text{Poisson}(\kappa S)$: the correct pivot under a fixed frontier is Poisson rather than $\text{Gamma}(M, 1)$, the correct pooled estimator is $(M - B)/S$ rather than $(M - 1)/S$, and confidence intervals are available in closed form without simulation-based inversion. The conditional likelihood costs one event per base, not one event in total—an exact analogue of the loss of one degree of freedom per group in the Neyman–Scott example. Distinguishing such schemes goes back to reliability theory; what is new here is the case of an undocumented frontier, where the exposure cannot be taken from a protocol and has to be reconstructed from the finds (Remark 5).
4. This gives $\kappa = 1.838$ with an exact 95% interval $[1.594, 2.109]$ and $p = 0.672$ against e^γ . The reported ten-percent excess over LPW—including our own in an earlier version of this work—is an artifact of the observation scheme. Under scheme B the old Gamma pivot rejects a true hypothesis in 11–16% of cases. Moving to scheme B also independently removes the systematic trend of the pooled estimate under truncation in $n_{\min}$.
5. The inference rests on a single assumption (A1): the frontier was chosen without regard to the finds. It cannot be checked from the data: $(M - 1)/S$ and $(M - B)/S$ are functions of the same statistics and differ by 6.9%. Without (A1) only the upper bound survives rigorously (Wald’s identity); the rule “stop after a long stretch without finds” biases even $(M - B)/S$ upwards, so there is no two-sided bracket. None of the violations of (A1) analysed here creates an excess over LPW. The four documented frontiers point, if anywhere, to a deviation towards stopping close to a find rather than “after a drought”; the hybrid estimate with the known frontiers for $b = 2$ and $b = 10$ is 1.853. The practical consequence

is the need to publish the search frontiers L_b (and, with them, the protocols of sieving completeness), as GIMPS does for $b = 2$. Until that is done, any pooled analysis of these sequences rests on an unverifiable assumption whose price is at least 7%, that is, twice the residual gap with LPW.

6. The LPW conjecture is not rejected by the available data, and base-dependent corrections are not required. But both of these are statements about the low sensitivity of the tests: the conditional multinomial homogeneity test resolves only a spread of κ_b of order $\pm 37\%$, and the pooled test only deviations of the constant of at least 21.3% (two-sided variant). Attaining a sensitivity of 10% would require the search frontiers to grow by a factor of about $5 \cdot 10^{19}$.
7. The LPW heuristic itself predicts at finite n an intensity $(\kappa/\ln b)(1 + c_b/t)$ with $c_b \in [\ln 2, 1.74]$, so comparing the constant-intensity estimate with e^γ is not a test of LPW. The second-order term is not determined by the data but shifts the estimate of κ by 8–16%; once it is taken into account, LPW is consistent with the data, the trend under truncation disappears, and the “slow convergence from below” is explained. Refining c_b is a problem for number theory; until it is solved, a test of LPW more precise than $\pm 10\%$ is out of reach.

A Verification of the source data

All twenty sequences (fifteen of the main set and five from the check of the selection limit) were cross-checked against the OEIS entries on 27 August 2026. Since only the triples $(N_b, n_{\min}, n_{\max})$ enter the pooled analysis, Table 10 gives, for the main set, the number of known primes—from which N_b is obtained by subtracting one—and both extreme indices; the full lists of indices are contained in the script `analysis_v2.py`.

Table 10. Cross-check against OEIS. The column “last addition” gives the provenance of the largest known index.

b	OEIS	primes	$n_{\min}$	$n_{\max}$	last addition
2	A000043	52	2	136 279 841	GIMPS, Oct. 2024
3	A028491	23	3	8 530 117	P. Bourdelais
5	A004061	19	3	3 300 593	P. Bourdelais
6	A004062	17	2	3 360 347	P. Bourdelais
7	A004063	10	5	1 264 699	P. Bourdelais
10	A004023	11	2	8 177 207	Propper and Batalov, May 2021
11	A005808	13	17	1 868 983	P. Bourdelais
12	A004064	14	2	769 543	P. Bourdelais, Dec. 2014
13	A016054	13	5	1 503 503	P. Bourdelais
14	A006032	11	3	1 724 417	P. Bourdelais, 13 Jul. 2026
15	A006033	10	3	639 833	P. Bourdelais, 22 Apr. 2019
17	A006034	12	3	1 990 523	P. Bourdelais, 3 Aug. 2020
18	A133857	8	2	1 270 141	P. Bourdelais
19	A006035	11	19	209 359	P. Bourdelais, 27 Aug. 2010
20	A127995	9	3	984 349	P. Bourdelais

A separate warning to the reader who checks the data against secondary sources: summary tables (the “Repunit primes” page in the OEIS Wiki, the article “List of repunit primes” in Wikipedia) are updated with a delay and, at the time of writing, lagged the OEIS entries themselves by one term for the bases $b = 14, 15, 17, 18, 20$. The OEIS entries should be treated as authoritative.

Independent recomputation of the prefixes

Cross-checking against OEIS confirms that the listed indices give primes, but it does not guarantee *completeness*: a missing term would shift N_b , and with it both pooled estimates. We closed this gap by direct recomputation. For each of the twenty bases $2 \leq b \leq 26$ that are not perfect powers, an independent implementation¹ tested *all* indices $n < 10^4$ consecutively, not only those listed in OEIS. Every negative verdict was recomputed by a second, independent backend (a blanket double-check, as in GIMPS) so as to rule out a lost find.

The result: 153 terms confirmed, none spurious, none missing, no disagreement between backends—across all twenty bases. Coverage is uneven across bases (from one term at $b = 18$ to twenty-two at $b = 2$): the main set $b \leq 20$ accounts for 128 confirmed indices out of 233 known. Above 10^4 a complete recomputation is computationally out of reach, and there we still rely on OEIS.

We note that this recomputation was not a formality: on the first run the independent implementation declared $R_{3181}(23)$ composite, disagreeing with A204940. Analysis showed an error in the implementation itself—a spike of FFT rounding error on the last iteration, after the last periodic check—and not in the OEIS data. After the fix no disagreements remained. We report this episode as an illustration of the thesis of §4: a false “composite” is invisible without recomputation, and that is exactly why frontiers and verification protocols should be published.

Declarations

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Competing interests

The author declares no competing interests.

Data and code availability

All sequences used are listed in Table 1.

Data. The script `fetch_data.sh` downloads the OEIS b-files for all fifteen sequences of the main set, and `fetch_extra.sh` does the same for the five bases $21 \leq b \leq 26$ used in the check of the selection limit (§2.2). The downloaded files are included in the repository, so the analysis reproduces without network access.

Analysis. The script `analysis_v2.py` (numpy, scipy, matplotlib); every number in this paper is tagged in its output (`results_v2.txt`) with a tag of the form [S3] identifying the corresponding section or table. The pooled analysis, all confidence intervals, the homogeneity test, the power

¹A hybrid CPU/GPU searcher: segmented sieving by small divisors, trial factoring in Montgomery arithmetic, PRP testing via GMP and GWNUM/IBDWT. The source code and the run logs are in the same repository, scripts `verify_sequences.sh` and `compare_verify.py`.

calculations and the structural models depend only on the triples $(N_b, n_{\min}, n_{\max})$; the full lists of indices are needed only for the $\hat{G}_b^{\text{OLS}}$ column of Table 5, for §7, for the truncation and for the variable-intensity models. The script reads them from the b-files in the `data/` directory (the loader is shared with `analysis.py` of version 1). The random number generator is initialized with a fixed seed (20260914), so all Monte Carlo quantities reproduce bit for bit. The script also produces all four figures (directory `figs_v2/`, with English labels in `figs_v2/en/`). Table 3 (stopping rules that violate (A1)) is produced by a separate script, `stopping_rules.py` (tag [S13], seed 20260924). The second-order term (§9), model (23) and the homogeneity test at $B = 20$ are computed by `lpw_second_order.py` (tags [S14]–[S17]), and the check of (A1) against the documented frontiers and the hybrid estimate by `frontiers.py` (tag [S18]; the sources of the frontiers are listed in the script itself). The output of all three scripts is saved in `results_v3.txt`.

Independent recomputation of the sequences. `verify_sequences.sh` runs the searcher over all indices $n < 10^4$ for each base, and `compare_verify.py` compares the result with the OEIS b-files, returning a non-zero exit code on any disagreement (Appendix A).

A caveat on the Monte Carlo quantities. The main result (20), the intervals of Table 5, Table 6 and Table 7 are *deterministic*: they are obtained by exact inversion of the Poisson distribution and do not depend on the random number generator. Only Tables 2, 4 and 3, the exact p -value of the homogeneity test and its power curve, and the null distributions of §7 depend on simulation; their accuracy is stated where they appear. This is an essential difference from the previous version, where the main interval estimate itself was simulation-based.

Repository: <https://github.com/Wilps93/repunit-hunt>, directory `paper/`. The file tree and the order of execution are described in `paper/README.md`.

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